Command Reference

LINE THERMAL PRINTER

MODEL: BL2-58
SP1-21
SK1-31 / 32/21/22/24 /
SP2-21 SP3-21 SD1-31
SK1-41 SD3-21 / 22
SK1-21H / 31H SK1-211 /
311 (NEW)

三栄電機株式会社

V5.3JR02 17.06

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Command Reference

Overview

1.1 Operation Mode

As control command, MODE A (ESC / POS compliant), describes a model-dependent commands, MODE B (only mode command).

Corresponding model are as follows.

MODE A: BL2-58 series SP1-21 series
SK1-31 / 32/21/22/24 series SP2-21 series SP3-21 series SD1-31 series
SK1-41 series SD3-21 / 22 series SK1-21 / 31H series SK1-211 / 311 series

MODE B: BL2-58 series SP1-21 series SD1-31 series

1.2 notation of numbers

Hexadecimal notation ... <number>	Example. <31>, <41>
Decimal notation ... the numbers	Example. 1, 100, 255
Representation of the binary numbers ... <number> B	Example. <Xxxxxxx0> B, <xxxxxxx1> B
Representation of the ASCII ... 'numbers'	Example. '0', '1'

1.3 Character Set

Character data sent from the host computer to the printer, alphanumeric and kana characters of 1 byte are all set, it is automatically converted to a full-width characters in symbol characters and double-byte. The contents of the character set, please refer to the "Character".

1.4 control command

Control command, starts the printing / end or line feed, a function of controlling the operation and modification functions of the paper feeding such as a printer. Character type selection, enlargement or formatting controls all functions related to such graphics printing.

1.5 transmission of printer

In the control command has the ability to control the transmission of data, such as notifying the state of the printer to the host. Matters shown below are details regarding this transmission function.

To execute, the input buffer during deployment can result in transmission delay on the states of the input buffer
There is.
· When sending is performed without confirmation of the host state.
- and valid for the virtual COM interface. Virtual COM is primarily serial interface,
USB (Communication Device Class), Bluetooth (SPP) is true. The other, in the SK1 series, in the release version V1.51 or later, USB (PRINTER DEVICE) also corresponding to the response function.

please note

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I. MODE A Command Reference (ESC / POS)

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

I-1. Paper feed command

CR

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] print return, line feed [code]

<0D>

Prints data [Function] print buffer, performs line feed on the basis of the line feed amount set. After [Detail]-execution, the print start position beginning.

- **CR** After **LF** It shall be void. - **LF** After **CR** It shall be valid.

LF

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] print return, line feed [code]

<0A>

Prints data [Function] print buffer, performs line feed on the basis of the line feed amount set. After [Detail]-execution, the print start position beginning.

- **CR** After **LF** It shall be void. - **LF** After **CR** It shall be valid.

FF

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] page length print (standard mode) / page memory print and return (page mode) [code] <0C>

In [Function] Standard mode, the page break based on the page length setting.

In the page mode, to return to the page memory batch after printing standard mode.

After [Detail]-run, the beginning and the next printing start position.

• After returning to the standard mode, **ESC S** The same state as.

ESC C

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name page length setting in the Code]

<1B> <43> n [domain] 1 ≦ n ≦ 255

[Function] to set the number of lines on a page. The execution of [Detail], page breaks, **FF**

Carried out by command.

• This command is valid only when the standard mode.

ESC J

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] printing and paper feed Code] <1B>
<4A> n [domain] 0 ≦ n ≦ 255

Prints data [Function] print buffer, performs paper feeding of the n × dot pitch. After [Detail]-execution, the print start position beginning.

- not affected by the amount of line feed setting.

· In the page mode, the positive y-axis movement of the page memory.

ESC j

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24	SP3-21	SD3-21 / 22	SK1-21H / 31H
	SK1-211 / 311					

[Name] printing and reverse paper feed Code] <1B>
<6A> n [domain] 0 ≦ n ≦ 255

Prints data [Function] print buffer, performs reverse paper feed of the n × dot pitch. Line feed operation when the [Detail] print buffer has data, the subsequent [n × dot pitch]

Perform a reverse direction paper feed.

When data is not present, it performs the reverse paper feed [n × dot pitch. · In the page mode, the cursor moves from the current value of the page memory in the opposite direction of the Y-axis. In the label mode, it discards the third parameter without running.

Note: 1. After the reverse direction paper feed, please refer to the printing operation after performing always 2mm or more positive direction paper feed.

Note 2: This command is, because it is designed for applications to adjust the starting position of printing (margin amount), uses other than that, the cause of the paper jam

Please note that the.

ESC d

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] Printing and n rows paper feed Code]
<1B> <64> n [domain] 0 ≦ n ≦ 255

And print data for [Function] print buffer, performs paper feeding of n rows. After [Detail]-run, the beginning and the next printing start position.

HT

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] Horizontal tab [code]

<09>

The [Function] printing position, moves to the next horizontal tab position.

If the Detail] and horizontal tab position is not set, the command is ignored.

And horizontal tab position may exceed the printing area is moved to the beginning of the line. Configuring horizontal tab position is carried out by ESC D. • The initial value of the horizontal tab, and every 8 characters.

ESC D

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] Horizontal tab settings [Code] <1B> <44>

n1 ... nk NULL [domain] $1 \leqq n \leqq 255$

$0 \leqq k \leqq 32$

[Function] Sets the horizontal tab position.

n indicates the number of digits to the set position from the head position of the line. k represents the number of data to be set.

Horizontal tab positions [Detail]-setting, and [character width × n].

· Character width, character spacing, lateral magnification is also included.

· Programmable tab positions and up to 32. If it exceeds 32, processed as normal data. · If you set the previous value smaller value during the configuration is recognized as NULL code. - Even if the character width is changed after setting, tab position set does not change.

ESC 2

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] initial line feed amount setting

Code] <1B> <32>

[Function] Sets the line feed amount per one row to an initial value.

ESC 3

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] line feed amount setting

Code] <1B> <33> n [domain] 0 ≦ n ≦

255

[Function] Sets the line feed amount per one row [n × dot pitch.

Both [details] Standard / page mode, can be set independent line feed.

• The initial line feed amount is set to n = 28.

• If a line of print height exceeds line feed amount, the print height is line feed. • In the case of a new line only,

according to the amount of line feed setting.

ESC SP

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] character right space amount setting of Code]

<1B> <20> n [domain] 0 ≦ n ≦ 127

[Function] the right amount of space of single-byte character set to [n × dot pitch]. [Detail]-right spacing

increases depending on the character lateral magnification.

- it does not affect the full-width characters.

- maximum value of settable right spacing is n = 127. • If more than the

maximum value, replacing the maximum value. • The initial value, and n = 0.

GS L

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] Set Left Margin [Code] <1D> <4C>

nl nh [domain] 0 ≦ nl ≦ 255

0 ≦ nh ≦ 255 n = nh ×

256 + nl

[Function] to set the left margin [n × dot pitch]. And valid only in the [Detail] Standard mode and the beginning of the line.

In the page mode is performed only setting. Maximum left margin settable, the horizontal printable area. • If more than the maximum value, replaced with the maximum value. • The initial value, and n = 0.

GS W

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] Set Print area width [Code] <1D>

<57> nl nh [domain] 0 ≦ nl ≦ 255

0 ≦ nh ≦ 255 n = nh ×

256 + nl

Set [Function] print area width [n × dot pitch. And valid only in the [Detail] Standard mode and the beginning of the line.

In the page mode is performed only setting.

· Programmable print area is horizontal printable area excluding the left margin. If you exceed it

It is rounded laterally printable area excluding the left margin. • The initial value (n) is, Paper width and printing width By different. (See table below)

Corresponding model	initial value	Print width / (dot)
SK1-41	831	104mm / (640)
SK1-41 / 31/32/311	639	80mm / (640dot)
SK1-41 / 31/32/311, SD1-31	575	72mm / (576dot)
SK1-31 / 32/21/22/24/311/211	447	56mm / (448dot)
SK1-31 / 32/21/22/24/311/211, SD3-22	431	54mm / (432dot)
BL2-58, SD3-21 / 22, SP1 / SP2 / SP3-21	383	48mm / (384dot)

ESC $

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] absolute positioning of the print area [Code]

<1B> <24> nl nh [domain] 0 ≦ nl ≦ 255

0 ≦ nh ≦ 255 0 ≦

nhnl ≦ 127

[Function] Sets the printing area in the absolute position relative to the left margin.

Setting width, and [(nh × 256 + nl) × dot pitch.

And valid only in the [Detail] Standard mode and the beginning of the line.

In the page mode is performed only setting. - exceeds the maximum

value set, to invalidate the command.

ESC a

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] justification [Code] <1B>

<61> n [domain] 0 ≦ n ≦ 2

[Function] align a line of print data to the designated position.

n = 0: Left n = 1:

centered n = 2: Right

And valid only in the [Detail] Standard mode and the beginning of the line.

In the page mode is performed only setting. · Set are performed

justification in the print area by. • The initial value, and n = 0.

ESC !

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] modified letter designation bulk Code]

<1B> <21> n [domain] 0 ≦ n ≦ 255

[Function] to specify the print mode at once.

BIT	Item Description	function
0	character font	0: Font A (12 × 24,24 × 24) 1: Font B (8 × 16,16 × 16)
1	Undefined	-
2	Undefined	-
3	highlighted characters	0: cancel 1: specified
4	double-height character	0: cancel 1: specified
5	double width characters	0: cancel 1: specified
6	Undefined	-
7	underline	0: cancel 1: specified

[Detail] · double height / lateral magnification specify both becomes to 4 double angle.

- underline amount is set to 2 dot pitch.
- it can also be set by another individual command, but to validate the command last processed. · Double-byte characters, character font, highlighted characters, and valid for the underline. - single-byte characters is valid for all the items. • The initial value, and n = 0.

[The specification which depend on the model] SP1-21

· Multiple of the thickness of the underline in the same row, to unify the thickest line.

ESC G

ESC E

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] specified and release of the highlighted

characters [code] <1B> <47> n

<1B> <45> n

[Domain] 0 ≦ n ≦ 255

[Function] to designate and release of the highlighted characters.

n = <xxxxxxx0> B: n = <xxxxxxx1> to

release B: to specify

And [Detail] · n least significant bit only valid.

• The initial value, and n = 0.

ESC {

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

Designation and canceling of [Name] Inverted

printing Code] <1B> <7B> n [domain] 0 ≦ n ≦ 255

[Function] to designate and release of the upside-down printing.

n = <xxxxxxx0> B: n = <xxxxxxx1> to

release B: to specify

And [Detail] · n least significant bit only valid.

- and Standard mode and only effective at the beginning of a line. In page mode, this command invalid. • The initial value, and n = 0.

ESC -

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

Designation and canceling of [Name] underline Code]

<1B> <2D> n [domain] 0 ≦ n ≦ 255

[Function] to designate and release of the underline.

n = <xxxxx000> B: underline 0 dot pitch

|

n = <xxxxx111> B: underline 7 dot pitch

And [Detail] · n the lower 3-bit only valid.

- only valid for single-byte characters.

- underline adds character width and with respect to its character space.

However, **HT** Not added to the part having been skipped by like. • Do not additions to the black-and-white inverted character. • The initial value, and n = 0.

[The specification which depend on the model] SP1-21

· Multiple of the thickness of the underline in the same row, to unify the thickest line.

GS!

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] set character size [Code] <1D>

<21> n [domain] 0 ≦ n ≦ 255

To specify the [Function] character size.

n = <xxxx0000> B: longitudinal magnification 1x <Min>

|

n = <xxxx0111> B: longitudinal magnification 8x <up>

n = <0000xxxx> B: lateral magnification 1x <Min>

|

n = <0111xxxx> B: lateral magnification 8x <up>

And valid for all characters except the [Detail] · HRI character.

· Magnification specification outside the specified range is ignored. •

The initial value, and n = 0.

GS B

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] specified and release of the black-and-white inverted

character [code] <1D> <42> n [domain] 0 ≦ n ≦ 255

[Function] to designate and release of the black-and-white inverted character.

n = <xxxxxxx0> B: n = <xxxxxxx1> to

release B: to specify

And [Detail] · n least significant bit only valid.

• The initial value, and n = 0.

GS b

Compatible models	SK1-41 / 31/32/21/22/24 SD3-21 / 22 SK1-21H / 31H SK1-211 / 311

[Name] specified and release of smoothing [code] <1D>

<62> n [domain] 0 ≦ n ≦ 255

[Function] to designate and release of font smoothing.

n = <xxxxxxx0> B: n = <xxxxxxx1> to

release B: to specify

And [Detail] · n least significant bit only valid.

• The initial value, and n = 0.

With respect to [apply for] · SK1-21 / 31 series, to apply from the firmware version 1.20 or later.

ESC 4

Compatible models	SK1-41 / 31/32/21/22/24 SD3-21 / 22 SK1-21H / 31H SK1-211 / 311

Specifying [Name] italics Code] <1B> <34>

To specify [Function] italic (italic characters). And valid for all characters except the

[Detail] · HRI character.

· Initial value is released.

[Applicable] · SK1-21 / 31 series: to apply to the later firmware version 1.98.

· SK1-41 / 24 series: to apply to the later firmware version 2.30.

ESC 5

Compatible models	SK1-41 / 31/32/21/22/24 SD3-21 / 22 SK1-21H / 31H SK1-211 / 311

Release of [Name] italics [code] <1B> <35>

To cancel the [Function] italics (italics). And valid for all characters except the [Detail] ·

HRI character.

· Initial value is released.

[Applicable] · SK1-21 / 31 series: to apply to the later firmware version 1.98.

· SK1-41 / 24 series: to apply to the later firmware version 2.30.

ESC _

Compatible models	SK1-41 / 24

[Name] specified and release of the upper line [code] <1B>

<5F> n [domain] 0 ≦ n ≦ 255

[Function] to designate and release of the upper line.

n = 0,48: Upper line release n = 1,49: Upper

line specified

[Detail] Upper line adds character width and with respect to its character space.

- However, **HT** Not added to the part having been skipped by like. • Do not additions to

the black-and-white inverted character. - width at the time of upper line specified, and

two dots. • The initial value, and n = 0.

[Applicable] · SK1-41 / 24 series: to apply to the later firmware version 2.26.

ESC M

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] selection of character font [Code] <1B>
<4D> n [domain] 0 ≦ n ≦ 255

[Function] to select the character font.

n = 0,48: Font A (12 × 24,24 × 24) n = 1,49: Font B (8 ×
16,16 × 16) n = 2,50: Font B (8 × 16,16 × 16) Note 1

[Detail] · This command is also valid for double-byte characters.

- **ESC !** But it is possible to set, but to validate the command last processed. • The initial value, and n = 0.

[Apply] · n = 2 is, BL2-58, SP1-21, SP2-21, SP3-21, not equipped to SD1-31.

ESC R

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] selection of international
character [code] <1B> <52> n [domain] 0
≦ n ≦ 8
Select Function] character set of the countries shown below.

n	Country
0	America
1	France
2	Germany
3	England
Four	Denmark
Five	Sweden
6	Italy
7	Spain
8	Japan

[Detail] · outside the specified range of data it will be ignored.

• The initial value, and n = 8.

ESC t

Compatible models	BL2-58	SP1-21	SP2-21 / SP3-21	SD1-31

[Name] selected character set table [Code] <1B> <74>

n [domain] 0 ≦ n ≦ 10

Select Function] character set shown below.

n = 0: PC437 / 1: Katakana / 2: PC850 / 3: PC852 / 4: PC857 / 5: PC858 / 6: PC863
7: PC865 / 8: PC866 / 9: WPC1252 / 10: PC860

[Detail] · outside the specified range of data it will be ignored.

- it does not affect the full-width character font.

· Not registered in the non-volatile memory, the volatile memory only rewritten. • In the factory setting, it is as follows.

Japanese model: initial value (n) is, and n = 1. Other than the

Japanese model: initial value (n) is the n = 0.

However, in the memory switch mounting machine, according to the setting of the memory switch. · Since the setting range is different by each model, see Technical Manual.

ESC t

Compatible models	SK1-41 / 31/32/21/22/24	SD3-21 / 22 SK1-21H / 31H SK1-211 / 311

[Name] selected character set table [Code] <1B> <74>

n [domain] 0 ≦ n ≦ 18

Select Function] character set shown below.

n = 0: PC437 / 1: Katakana / 2: PC850 / 3: PC852 / 4: PC857 / 5: PC858 / 6: PC863
7: PC865 / 8: PC866 / 9: WPC1252 / 10: PC860 / 11: WPC1252-2 / 12: PC862 13: WPC1254 / 14: WPC1250 / 15: WPC1251 / 16: PC864 / 17: Reserve / 18: PC737

[Detail] · outside the specified range of data it will be ignored.

- it does not affect the full-width character font.

· Not registered in the non-volatile memory, the volatile memory only rewritten. • In the factory setting, it is as follows.

Japanese model: initial value (n) is, and n = 1. Other than the

Japanese model: initial value (n) is the n = 0.

However, in the memory switch mounting machine, according to the setting of the memory switch. · Since the setting range is different by each model, see Technical Manual.

In [Applicable] · SK1-21 / 31, the parameters of the 0 ≦ n ≦ 15 applies the release version V1.75 or later.

In · SK1-21 / 31, parameters n = 16 applies the release version V1.98 or later. In · SK1-41 / 24, parameters n = 16 applies the release version V2.30 or later. In · SK1-21 / 31, parameters n = 18 applies the release version V1.99 or later. In · SK1-41 / 24, applied to the parameters of 0 ≦ n ≦ 16.

ESC &

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] registration of download characters

[Code] <1B> <26> y c1 c2 [x1 d1 ... d (y × x1)] 1 ... [xk d1 ... d (y × xk)] cn [domain] y = 3

20h ≦ c1 ≦ c2 ≦ 7Eh, cn = (c2-c1) +1 0 ≦ x ≦ 12
(Font A when selected) 0 ≦ x ≦ 9 (Font B when
selected) 0 ≦ d ≦ 255

Define the download pattern [Function] specified character code.

y = Exit code x = number of lateral bit
vertical number of bytes c1 = start
character definition code c2 = character
definition

And the case of the [Detail] · one character only of the definition c1 = c2.

· D is the download character of graphic data. And right space left over by the designation of x is treated as a blank. • If the last time you specified in the registered code, the process by overwriting. • If you want to validate the registration character fonts, **ESC %** Setting is required. · Font B is selected, the horizontal-vertical (8 × 16) and the output of the dot.

Registration Image

Font 12 × 24

d1	d4	...	d34
d2	d5	...	d35
d3	d6	...	d36

Font 8 × 16

d1	d4	...	d25
d2	d5	...	d26
d3	d6	...	d27

MSB

LSB

Character output range

Font 12 × 24

d1	d4	...	d34
d2	d5	...	d35
d3	d6	...	d36

Font 8 × 16

d1	d4	...	d22	
d2	d5	...	d23	
		...		

ESC ?

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] cancellation of download characters [code]

<1B> <3F> n [domain] 20h ≦ n ≦ 7Eh

[Function] deletes the specified code download character. [Detail] · n represents the character code that you define.

After the erasure is to print the internal character.

• If the specified character code is undefined, the command is ignored.

ESC %

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] specified and release of download characters [code]

<1B> <25> n [domain] 0 ≦ n ≦ 255

[Function] to designate and release of the download character set.

n = <xxxxxxx0> B: n = <xxxxxxx1> to

release B: to specify

And [Detail] · n least significant bit only valid.

• If you download to release the character set, specify the internal character set. • If you download you specify the character set, specify the download character set. · Undefined code specifies the internal character set. • The initial value, and n = 0.

ESC *

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] bit image specified Code] <1B> <2A> m nl nh [d1 ...

dk] [domain] m = 0,1,32,33

0 ≦ nl ≦ 255 0 ≦

nh ≦ 3 0 ≦ d ≦

255

[Function] nl, the number of dots specified by nh, specifies the bit image mode m.

m	mode	Longitudinal direction The number of dots	The number of horizontal direction dot	The number of data (k)
0	8 dot single density	8	Refer to the table below.	nh × 256 + nl
1	8-dot double-density	8		nh × 256 + nl
32	24 dot single density	twenty four		(Nh × 256 + nl) × 3
33	24-dot double-density	twenty four		(Nh × 256 + nl) × 3

[Detail] · m if out of the range, the process data of subsequent nl as normal data.

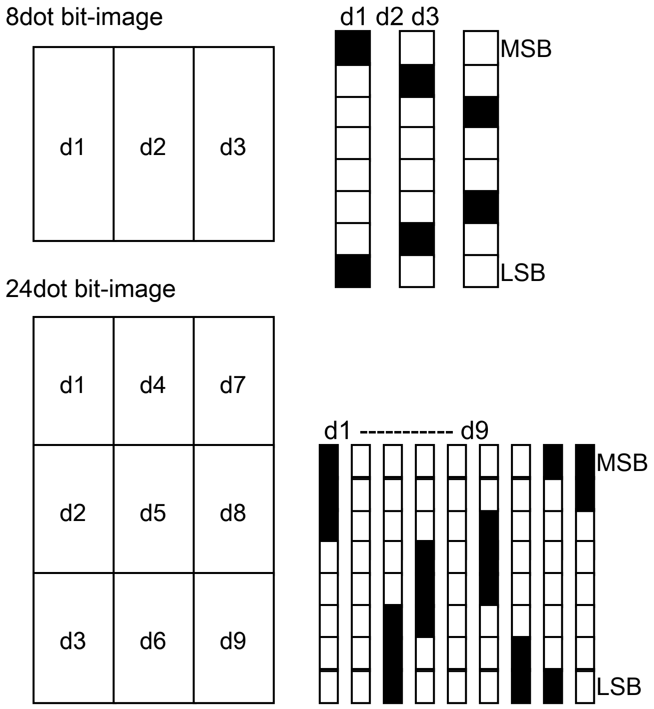
· Nl, nh indicates the number of lateral dots of the bit image to be printed. • If you make a dot specified in the printable area outside, it discards the data. The data deployed position, according to the mapping start position of the time.

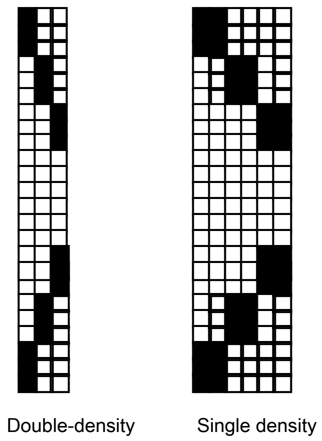
· Influence of upside-down printing is subject. Others (double, stressed, black-and-white reversal, etc.) is not affected by the. · How to Deploy in the page mode, **ESC L** checking ... And deployment image of bit image data, refer to the illustration on the next page.

Table (the number of horizontal direction dot):

Print width / (dot)	Single density / double density	Corresponding model
104mm (832)	416/832	SK1-41
80mm / (640dot)	320/640	SK1-41 / 31/32 / 31H / 311
72mm / (576dot)	288/576	SK1-41 / 31/32 / 31H / 311, SD1-31
56mm / (448dot)	224/448	SK1-31 / 32/21/22/24 / 21H / 31H / 211/311
54mm / (432dot)	216/432	SK1-31 / 32/21/22/24 / 21H / 31H / 211/311 SD3-22
48mm / (384dot)	192/384	BL2-58, SD3-21 / 22, SP1 / SP2 / SP3-21

Expand Image





Sample Code Example] SK1-21,8 bit single density, lateral 80 dots specified

```
n = 0;
buf [n ++] = 0x1B; buf [n
++] = 0x2A; buf [n ++] =
0x00; buf [n ++] = 0x50;
buf [n ++] = 0x00;


for (i = 0; i <10; i ++) {
     buf [n ++] = 0x88; buf [n
    ++] = 0x44; buf [n ++] =
    0x22; buf [n ++] = 0x11;
    buf [n ++] = 0x11; buf [n
    ++] = 0x22; buf [n ++] =
    0x44; buf [n ++] = 0x88;}


Ret = WriteFile (hPort, buf, n, & dwSendSize, NULL);
```

[Print result Samples

GS *

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] Download bit image registration [Code] <1D> <2A> xy

[d1 ... d (x × y × 8)] [domain] 1 ≦ x ≦ 255

1 ≦ y ≦ 48 However (x × y × 8) ≦ user free memory capacity

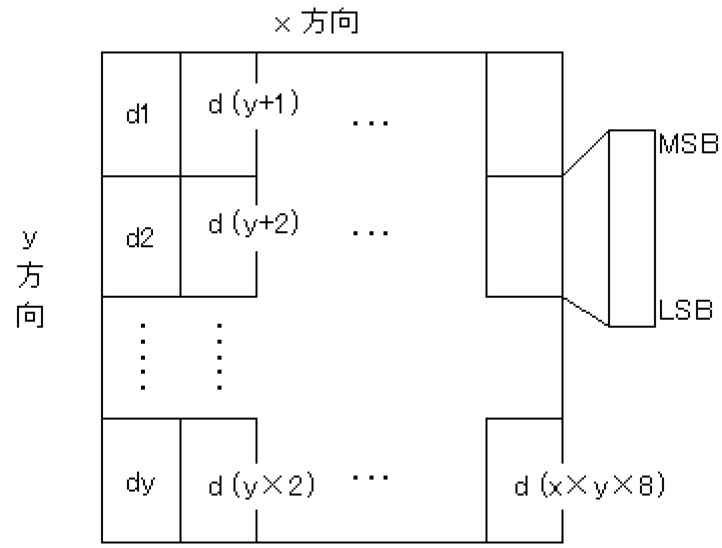
0 ≦ d ≦ 255

[Function] x, define a download bit image dot number specified by y.

x is the horizontal direction (x × 8) to specify the number of dots. y is the vertical direction (y × 8) to specify the number of dots.

[Detail] · outside the specified range of data it will be ignored.

• For user memory, refer to the separate "Technical Manual". And deployment method is shown in the figure below.



GS /

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] Download bit image printing [Code] <1D> <2F> m

[domain] 0 ≦ m ≦ 3,48 ≦ m ≦ 51

[Function] In the given mode m, to print the downloaded bit image.

m	Print mode	Content
0,48	Normal mode	It is printed in the normal magnification
1,49	Double width mode	Printed in the lateral magnification
2,50	Double-height mode	Printing in the double-height
3,51	4-fold mode	4 times to be printed

If the [Detail] · download bit image has not been defined, the command is ignored.

Print If buffer has data, after printing it, download bit

Carry out the printing of the image. Not affected by the print mode with the exception of (standard mode), upside-down printing. How to deploy by the page mode, **ESC L** checking ... · In printable area outside to print the fractional part in the right direction bytes.

Sample Code example] SK1-21, data size vertical and horizontal 8 dots, print mode specifies the normal mode

```
n = 0;
buf [n ++] = 0x1D; buf [n
++] = 0x2A; buf [n ++] =
0x08; buf [n ++] = 0x08;


for (i = 0; i <64; i ++) {
     buf [n ++] = 0xFF; buf [n
    ++] = 0x00; buf [n ++] =
    0xFF; buf [n ++] = 0x00;
    buf [n ++] = 0xFF; buf [n
    ++] = 0x00; buf [n ++] =
    0xFF; buf [n ++] = 0x00;}


buf [n ++] = 0x1D; buf [n
++] = 0x2F; buf [n ++] =
0x00;
Ret = WriteFile (hPort, buf, n, & dwSendSize, NULL);
```

[Print result Samples

DC2 V

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

Printing [Name] raster bit image Code] <12> <56> nl nh

[d1 ... dk] [domain] 0 ≦ nl ≦ 255

0 ≦ nh ≦ 255 0 ≦

d ≦ 255

[Function] nl, specifies the bit image a specified number of lines in nh.

1 line data number (m)	The entire number of data (k)
Refer to the table below.	(Nh × 256 + nl) × m

[Detail] · nl, nh indicates the number of vertical lines.

Page mode, to this command and invalid. * 1

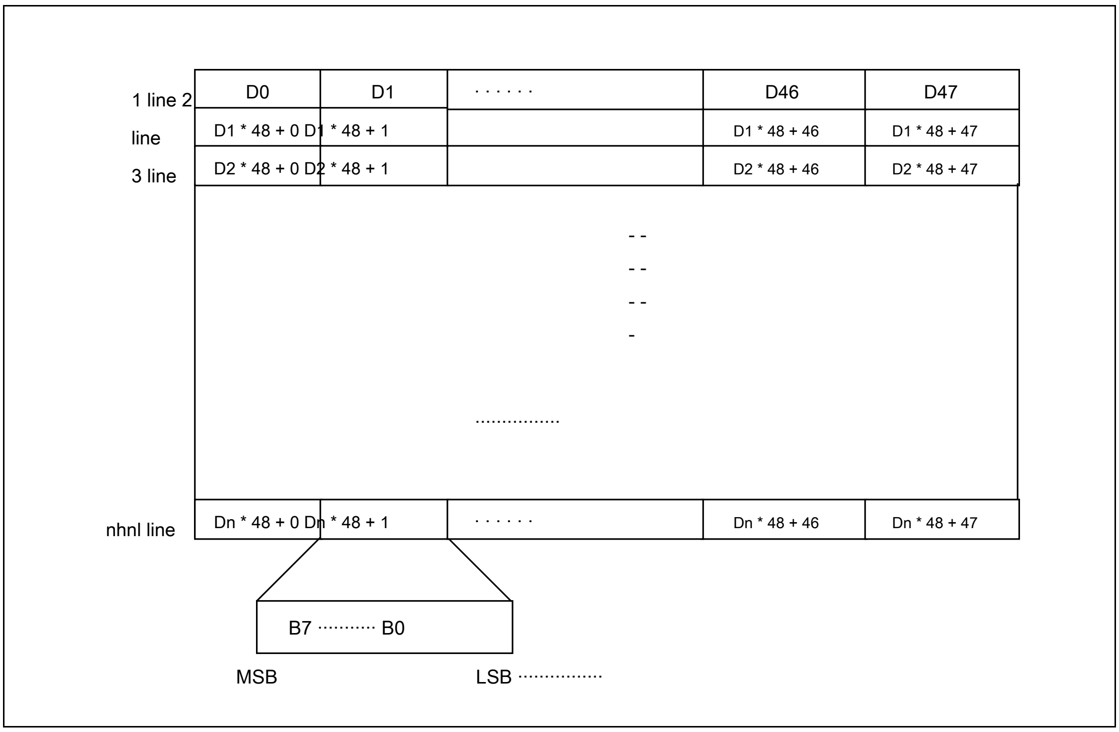
• Specifying and release command of the upside-down printing in this command is invalid. And deployment image is shown in the illustrative example.

*** 1. In the SK1 series, and apply from the release version V1.51 and later to the page commands (positive direction). SD3-21 / 22 series, apply than the initial release version of the page commands (positive direction).**

Table (1 line number data):

Print width / (dot)	Data number (m)	Corresponding model
104mm / (832dot)	104	SK1-41
80mm / (640dot)	80	SK1-41 / 31/32 / 31H / 311
72mm / (576dot)	72	SK1-41 / 31/32 / 31H / 311, SD1-31
56mm / (448dot)	56	SK1-31 / 32/21/22/24 / 21H / 31H / 211/311
54mm / (432dot)	54	SK1-31 / 32/21/22/24 / 21H / 31H / 211/311 SD3-22
48mm / (384dot)	48	BL2-58, SD3-21 / 22, SP1 / SP2 / SP3-21

Illustrative example: m = case 48



Sample Code Example] SK1-21, print width 54 mm, the vertical line 8 dots specified

```
n = 0;
buf [n ++] = 0x12; buf [n
++] = 0x56; buf [n ++] =
0x08; buf [n ++] = 0x00;


for (i = 0; i <8; i ++) {
        for (j = 0; j <27; j ++) {
                buf [n ++] = 0xFF; buf [n
        ++] = 0x00;}}


Ret = WriteFile (hPort, buf, n, & dwSendSize, NULL);
```

[Print result Samples

DC2 v

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

Printing of raster bit image by [Name] compression

[Code] <12> <76> n [m1 [code + Data Length] [d1 ... dk]] ... [mn [code + Data Length] [d1 ... dk]] [domain] 0 ≦ n ≦ 255

0 ≦ m ≦ 3 0 ≦ d

≦ 255

The [Function] compression mode one line specified by m, the bit performs image development of the n lines

To print the image.

n: Specifies the number of lines to expand the bit image [n × dot line] m: specifies the compression mode

m = 0: Specify normal compression

m = 1: Blank specified line (all 1 line and 0) m = 2: Copy the last specified line.

m = 3: Copy the last time specified line, specify the override to a specific byte position.

[Compression rule: m = 0]

· Adopted a byte length in the compression mode, be sure to head to specify the sign + data length]. - The sign specifies the uncompressed or compressed. It shows the compression method, the following

1. The specified compression

   Sign + data length: (80H: sign) + (0 ~ 127 (7FH): data length ① ※ see Table 1 below) data: 1 byte only specified, the image deployment in succession the specified data length.

2. uncompressed

   Sign + data length: (00H: sign) + (1 ~ 127 (7FH): data length ② ※ Table 1 below reference) data: data length, to specify the image data to image deployment.

Table (1 line number data):

Print width / (dot)	Data length ①	data length ②	Corresponding model
104mm / (832dot)	0-103	1-104	SK1-41
80mm / (640dot)	0-79	1-80	SK1-41 / 31/32 / 31H / 311
72mm / (576dot)	0-71	1-72	SK1-41 / 31/32 / 31H / 311, SD1-31
56mm / (448dot)	0-55	1-56	SK1-31 / 32/21/22/24 / 21H / 31H / 211/311
54mm / (432dot)	0-53	1-54	SK1-31 / 32/21/22/24 / 21H / 31H / 211/311 SD3-22
48mm / (384dot)	0-47	1-48	BL2-58, SD3-21 / 22 / SP1 / SP2 / SP3-21

※ 1. The total value of all of the data length, please be configured to match the number of data.

Example 2) m = 0, 0x89 0xFF 0x46 AA (1) BB (2) ··· EE (70)

① the compressed image data <FF>, to place 10 bytes continuously. ② non-compressed image data <AA, BB, CC, DD, EE, FF> a, to place the 0x46 (70D) bytes.

Compressed designated data length ① 10 + uncompressed data length ② 70 = Number of data 80

[Compression rule: m = 3]

• In this mode, first copy the previous line, the data that you want to overwrite,

To specify the deployed position] + [data].

1. compression specify the deployed

position: 0-127

Data: 1 byte only specify placing directly the image data to the deployed position. End code: 80H ~ FFH <that the most significant bit is 1>

Example) m = 3, 0A AA 10 BB 80

① Copy the last line, the D (10) to <AA> / D (16) to override the <BB>.

To invalidate the designation of [Detail], upside-down printing.

ESC b

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP3-21	SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311					

[Name] width printing of the specified raster bit image Code] <1B> <62>

y nl nh [d1 ... dk] [domain] 0 ≦ y ≦ table: see

0 ≦ nl ≦ 255 0 ≦

nh ≦ 255 0 ≦ d ≦

255

Specified in [Function] width y, nl, specifies the bit image a specified number of lines in nh. [Detail] · nl, nh indicates the number of vertical lines.

· Y indicates the number of bytes in width.

• Specifying and release command of the upside-down printing in this command is invalid. And deployment images, **DC2 V** See illustration page.

Maximum printing width / (dot)	y maximum value	Corresponding model
104mm / (832dot)	104	SK1-41
80mm / (640dot)	80	SK1-41 / 31/32 / 31H / 311
72mm / (576dot)	72	SK1-41 / 31/32 / 31H / 311, SD1-31
56mm / (448dot)	56	SK1-31 / 32/21/22/24 / 21H / 31H / 211/311
54mm / (432dot)	54	SK1-31 / 32/21/22/24 / 21H / 31H / 211/311 SD3-22
48mm / (384dot)	48	BL2-58, SD3-21 / 22, SP1 / SP2 / SP3-21

[Applicable] · SK1-21 / 31: to apply to the release version 1.20 or later.

· BL2-58: to apply to the release version 1.40 or later. · SP1-21: to apply to the release version 1.40 or later.

Sample Code Example] SK1-21, width 26mm specified, the vertical line 8 dots specified

```
n = 0;
buf [n ++] = 0x1B; buf [n
++] = 0x62; buf [n ++] =
0x1A; buf [n ++] = 0x08;
buf [n ++] = 0x00;


for (i = 0; i <8; i ++) {
        for (j = 0; j <13; j ++) {
                buf [n ++] = 0x80; buf [n
        ++] = 0x08;}}


Ret = WriteFile (hPort, buf, n, & dwSendSize, NULL);
```

[Print result Samples

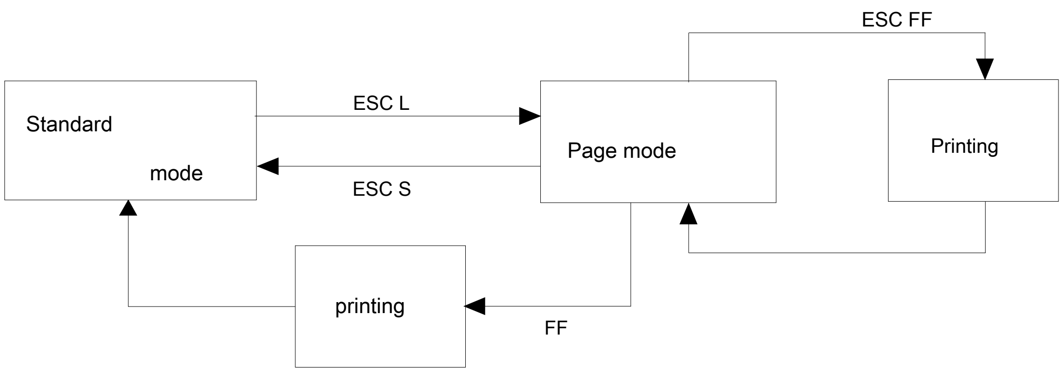
I-7. Page mode command

Description of the page command

To this printer, the print mode There are two types of standard mode and page mode. Standard mode (when the power is turned on, starting with this mode), print command (**CR** , **LF** Etc.) is a mode for performing a printing operation in time to receive. Page mode, printing instruction (**CR** , **LF** Is also received etc.) does not perform a printing operation, it performs a write to the area of the page memory, **ESC FF** Or **FF** By the instructions, you can make printing operation in a batch area of the page memory.

When you described in the operation example, in the standard mode, "SANEI" when sending the data that <CR>, "SANEI" and will perform a one-line printing, in the page mode, printing of "SANEI" is not carried out, page written and "SANEI" in memory, will be deployed position in the memory is moved.

Relationship between the page mode and standard mode is as follows.



ESC L

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP3-21	SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311					

[Name] page mode selection [code] <1B>

<4C>

[Function] switch from the standard mode to the page mode. And valid only in the [Detail]

Standard mode and the beginning of the line.

- **FF** Or **ESC S** By returning to the standard mode. · Character deployed position **ESC W** In specified. · Character development direction **ESC T** In specified.

· The following commands, in the page mode and standard mode, with independent settings.

① amount of space set: **ESC SP, FS S**

② line feed amount setting : **ESC 2, ESC 3**

· The following commands, but you can set the value in the page mode, the printing of the page mode It becomes effective at the time of standard mode return without being reflected. ① print area specified: **GS L, GS W, ESC $**

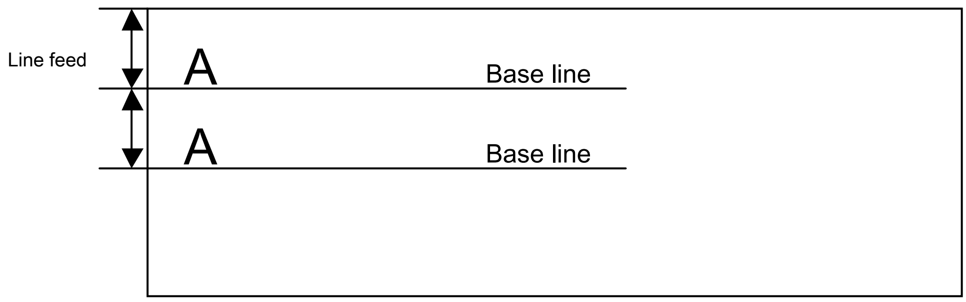
② justification : **ESC a**

· The following command will be ignored in the page mode.

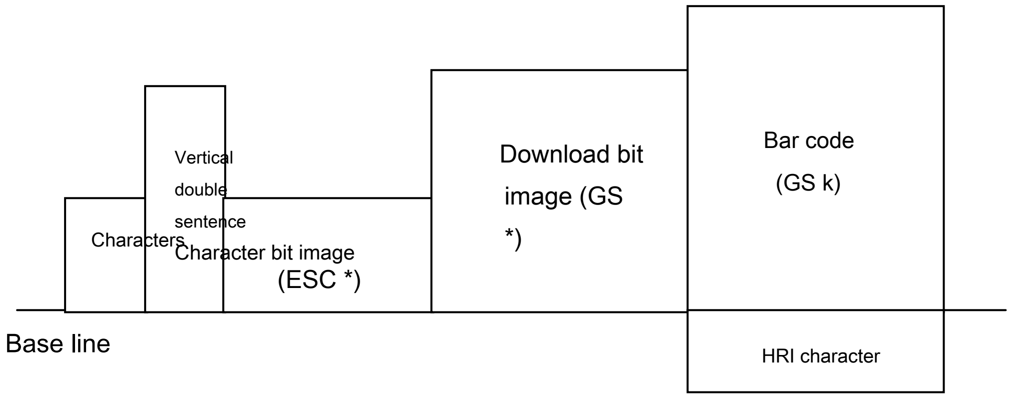
① upside-down printing: **ESC {**

- **ESC @** , In order to perform the initialization of each mode, to return to the standard mode.

<Deployment in the page mode>



The deployed position of the character data



The deployed position of the print data

ESC S

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP3-21	SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311					

[Name] Standard mode selection [code] <1B> <53>

Switch to standard mode from the [Function] page mode. Only effective when processing the [Detail]-page mode.

- Even to the end without being printed in the case of the page memory there is data. • After this command is executed, the print start position of the beginning of the line.

ESC FF

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP3-21	SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311					

[Name] page batch printout of memory [code] <1B>
<0C>
In the [Function] page mode perform batch printing of the printing area. Only effective when processing the [Detail]-page mode.

• Even after execution **ESC W** , **ESC T** I want to preserve the settings of. • After the execution also to hold the data of the page memory. (Except SK1-41)

[The specification which depend on the model]

In · SK1-41, data of the page memory after execution disappears.

CAN

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP3-21	SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311					

[Name] Clear of the print buffer (standard mode)

Clear of the page memory area (page mode)

[Code] <18>

[Function] it is clear of the print buffer in the standard mode.

In the page mode **ESC W** Collectively clear the print area that is specified.

After executing the [Detail] Standard mode, the print start position beginning.

After the execution in page mode, **ESC T** Back to the deployment start position.

ESC W

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP3-21	SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311					

[Name] setting of the expansion area

[Code] <1B> <57> xl xh yl yh dxl dxh dyl dyh [domain] 0 ≦ (xh × 256 + xl) ≦

See table below

1 ≦ (dxh × 256 + dxl) ≦ See table below

Print width	X maximum value	Dx maximum value	Corresponding model
104mm / (832dot)	830	831	SK1-41
80mm / (640dot)	638	639	SK1-41 / 31/32 / 31H / 311
72mm / (576dot)	574	575	SK1-41 / 31/32 / 31H / 311, SD1-31
56mm / (448dot)	446	447	SK1-31 / 32/21/22/24 / 21H / 211/311
54mm / (432dot)	430	431	SK1-31 / 32/21/22/24 / 21H / 211/311 SD3-22
48mm / (384dot)	382	383	BL2-58, SD3-21 / 22, SP1 / SP3-21

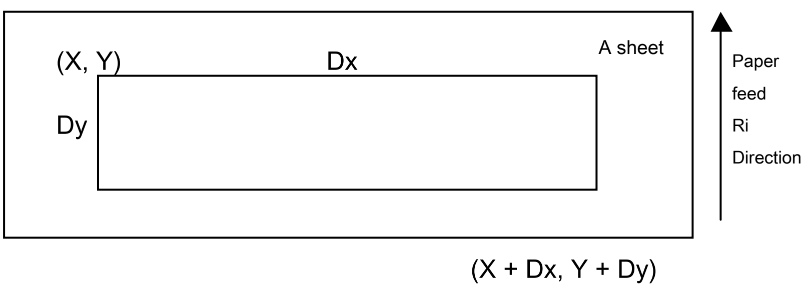
0 ≦ (yh × 256 + yl) ≦ See table below 1 ≦ (dyh × 256 + dyl) ≦ See table below

Page length	Y maximum value	Dy maximum value	Corresponding model
350mm (2800dot)	2798	2799	SK1-41 / 211/311
300mm (2400dot)	2398	2399	SK1-31 / 32 (printing width 72mm or less) * 1 SK1-21 / 22/24
250mm (2000dot)	1998	1999	SK1-31 / 32 (printing width 80mm or less) SK1-21H / 31H
200mm (2000dot)	1598	1599	SD3-21 / 22
116mm (928dot)	926	927	SP1 / SP3-21
85mm (680dot)	678	679	SD1-31
60mm (480dot)	478	479	BL2-58

* 1. In SK1-31 / 32/21/22, it can cope from the release version V1.54 or later.

To set the print area in the [Function] page mode.

①X axis origin (X) = (xh × 256 + xl) × dot pitch ②Y axis origin (Y) = (yh × 256 + yl) × dot pitch ③X axis length (Dx) = (dxh × 256 + dxl) × dot pitch ④Y axis length (Dy) = (dyh × 256 + dyl) × dot pitch



Only effective when processing the [Detail]-page mode.

· The way, also transmit the parameters of the definition outside, and disabled to get up to dyh command. And deployment direction of the character position, the starting point **ESC T** In specified. · X-direction, exceed the maximum value of the Y-direction, it replaces the maximum value. · When the page for printing is a printing length a set Y-axis maximum value. • If not performed setting determines the printing length by the initial setting. - a new line from the base line, in accordance with the amount of line feed setting. • The initial value is in accordance with the maximum value of the Page Setup. • The maximum expand the number of digits of the character, to 200 digits.

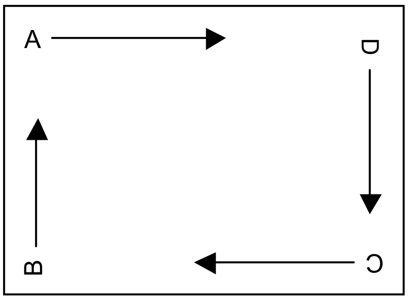
ESC T

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP3-21	SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311					

[Name] Selection of the print direction and the starting point [Code] <1B> <54> n [domain] 0 ≦ n ≦ 3

Selecting a printing direction and the starting point of the character in the [Function page mode.

n	Starting point and developing direction
0	A
1	B
2	C
3	D



If treatment with [Detail]-page mode If only be effective.

• Print deployed position, **ESC W** In a designated print area. · The developing direction, the adjustment of the X-axis / Y-axis differ.

① expansion direction (A, C)

Y-axis: ESC J, ESC 2, ESC 3 X axis: ESC SP, FS S ② expansion direction (B, D)

Y-axis: ESC SP, FS SX axis: ESC J, ESC 2, ESC 3 · initial value is n = 0.

ESC PC

Compatible models	SK1-41

- -

The definition of the string format to the [Name] page memory

[Code] <1B> <50> <43> n1 n2 n3 <3B> x1 x2 x3 x4 <2C> y1 y2 y3 y4 <2C>
w <2C> h <2C> c <2C> r1 r2 <2C> d1 d2 <LF> <00>

[Domain] '0' ≦ n1 n2 n3 ≦ '9' ("000" ≦ n1n2n3 ≦ "199")
'0' ≦ x1 x2 x3 x4 ≦ '9' '0' ≦ y1 y2 y3 y4 ≦
'9' '1' ≦ w ≦ '8' '1' ≦ h ≦ '8' '0' ≦ c ≦ '3 " 0
'≦ r1 ≦' 3 " 0 '≦ r2 ≦' 3 " 00 '≦ d1d2 ≦
'63'

[Function] defines the format information of the character string data to be direct to the rendering to the page memory.
It lists the parameters for the detail below. n1n2n3

	: String number
x1x2x3x4	: The starting point X-axis position Unit 0.1mm
y1y2y3y4	: Starting Y-axis position units 0.1mm
x	: Ratio of character width (1, 2, 3, 4, 5, 6, 7, 8-fold)
h	: Character height ratio (1, 2, 3, 4, 5, 6, 7, 8-fold)
c	: Character font (0: Font-A (12 × 24,24 × 24), 1: Font-B (8 × 16,16 × 16), 2: Font-A (12 × 24,24 × 24) + highlighted characters, 3: Font-B (8 × 16,16 × 16) + highlighted characters)
r1	: Direction of rotation of the character (0: 0 degrees / l: 90 ° / 2: 180 degrees / 3: 270 degrees)
r2	: Direction of rotation of the string (0: 0 degrees / l: 90 ° / 2: 180 degrees / 3: 270 degrees)
d1d2	: Character pitch

Only effective when processing the [Detail]-page mode.

- the starting point X, Y is, starting from the upper left corner of the first character. And information of the character string is specified by the ESC RC command.

• This command is not the concept of the base line, to reflect the information of the string to the page memory In order for a command, ESC W, not affected by the T command. • This command is a preliminary specification, because they consider making the extension of future specifications, please consideration so as not to be affected after expansion to protect the specified value of the parameter.

ESC RC

Compatible models	SK1-41

The definition of the character string data to the [Name] page memory

[Code] <1B> <52> <43> n1 n2 n3 <3B> d1 d2 ... dn LF <00>

[Domain] '0' ≦ n1 n2 n3 ≦ '9' ("000" ≦ n1n2n3 ≦ "199")

32 ≦ d1 d2 ... dn ≦ 255

[Function] defines the character string data to be direct to the rendering to the page memory.

It lists the parameters for the detail below. n1n2n3

: String number

a1a2 ... an : Character string data (up to 100 bytes)

Only effective when processing the [Detail]-page mode.

· Character string data registered draws at runtime ESC FF or FF command. - single-byte characters are treated as one-byte character string data, double-byte characters are treated as 2 bytes. • This command is not the concept of the base line, to reflect the information of the string to the page memory

In order for a command, ESC W, not affected by the T command. • If you want to register a double-byte characters, set the Shift-JIS in the pre-FS C command.

ESC X

Compatible models	SK1-41

[Name] Clear the character string data [code]

<1B> <58> LF <00>

[Function] to clear all of the character string data defined in the ESC RC command. Only effective when processing the [Detail]-page mode.

- string format is not clear.

ESC Z

Compatible models	SK1-41

[Name] string format, string clear border format of the clear [code] of data <1B> <5A> LF <00>

[Function] ESC string format defined in the PC command, was defined by the ESC RC command

All the borders format defined in the character string data Oyobi ESC I command to clear.

Only effective when processing the [Detail]-page mode.

ESC e

Compatible models	SK1-41

- -

[Name] Clear Border Data Code] <1B> <65> n1 n2 n3 LF

<00> [domain] '0' ≦ n1 n2 n3 ≦ '9'

("000" ≦ n1n2n3 ≦ "199")

To clear the specified border format in the [Function] n1n2n3. Only effective when processing the [Detail]-page mode.

ESC l

Compatible models	SK1-41

- -

[Name] definition of borders data

[Code] <1B> <6C> n1 n2 n3 <3B> x1 x2 x3 x4 <2C> y1 y2 y3 y4 <2C>
dx1 dx2 dx3 dx4 <2C> dy1 dy2 dy3 dy4 <2C> d <2C> w <LF> <00>

[Domain] '0' ≦ n1 n2 n3 ≦ '9' ("000" ≦ n1n2n3 ≦ "099")

'0' ≦ x1 x2 x3 x4 ≦ '9' '0' ≦ y1 y2 y3 y4 ≦ '9' '0'
≦ dx1 dx2 dx3 dx4 ≦ '9' '0' ≦ dy1 dy2 dy3 dy4
≦ '9' '0 '≦ d ≦' 2 " 1 '≦ w ≦' 9 '

[Function] to define the borders data to the rendering to the page memory to direct.

It is deployed directly to the page memory in accordance with the present borders data. n1n2n3

: Number of line

x1x2x3x4 : The starting point X-axis position Unit 0.1mm

y1y2y3y4 : Starting Y-axis position units 0.1mm

Dx1dx2dx3dx4: end point X-axis position units 0.1mm

dy1dy2dy3dy4: ending Y-axis position units 0.1 mm d

: Line direction (0: horizontal lines / 1: vertical line / 2: frame)

w : The number of line width dot (1 dot = 0.125mm)

Only effective when processing the [Detail]-page mode.

y direction and horizontal lines (d = '0') specifies the equivalence (y1y2y3y4 = dy1dy2dy3dy4). x-direction and vertical lines (d = '1') designates equivalence (x1x2x3x4 = dx1dx2dx3dx4). - it is not possible to draw a diagonal line in this command.

• This command has no concept of the base line, to be reflected directly to the page memory. · ESC W, not affected by the T command.

• This command is a preliminary specification, because they consider making the extension of future specifications, please consideration so as not to be affected after expansion to protect the specified value of the parameter.

<Actual borders print the relationship between the print position specification>

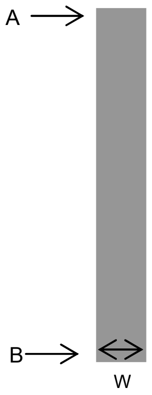
A : Print start position (x1x2x3x4, y1y2y3y4) B

: Print end position (dx1dx2dx3dx4, dy1dy2dy3dy4) W

: The width of the border

(1) Horizontal line (d = '0')

(2) Vertical line (d = '1')



(3) Frame (d = '2')

ESC =

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] selection of peripherals Code]

<1B> <3D> n [domain] 0 ≦ n ≦ 255

[Function] data from the host computer to select a valid peripheral device.

BIT Function 0
0: Printer disabled 1: Printer Enabled 1 Undefined 2
Undefined 3 Undefined 4
Undefined 5 Undefined 6
Undefined 7 Undefined

If the Detail] printer is disabled selected, printer printer by the command from the following data

Data but received until the effective selected again (**ESC =** Read and discarded the excluded).

ESC i

Compatible models	SP1-21	SK1-41 / 31/32/21/22/24 SD3-22		SK1-21H / 31H SK1-211 / 311

[Name] full cut [code] <1B> <69>

Do a full cut of the [Function] paper.

ESC m

Compatible models	SP1-21	SK1-41 / 31/32/21/22/24 SD3-22		SK1-21H / 31H SK1-211 / 311

[Name] partial cut [code] <1B> <6D>

Carry out the [Function] paper partial cut (leaving one point).

GS V

Compatible models	SP1-21	SK1-41 / 31/32/21/22/24 SD3-22		SK1-21H / 31H SK1-211 / 311

[Name] Paper cut Code] <1D> <56>

m

<1D> <56> mn (m = 65, 66)

[Domain] m = 0,1,48,49,65,66

$0 \leqq n \leqq 255$

[Function] the execution of the specified paper cut.

m = 0,48: performing the full cut. m = 1,49: performing

a partial cut.

m = 65: perform full-cut after performing the paper feeding [n × dot pitch. m = 66: performing a partial cut after performing the paper feeding [n × dot pitch.

ESC c 3

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

To [Name] PE signal output, choose a valid paper detector Code] <1B> <63> <33> n [domain] $0 \leqq n \leqq 255$

[Function] selecting a detector for detecting a paper-out signal.

BIT function 0	
Note 1	0: roll paper near-end detector disabled 1: roll paper near-end detector enabled 1
Note 1	0: roll paper near-end detector disabled 1: roll paper near-end detector enable 2
	0: roll paper end detector disabled 1: roll paper end detector enable 3
	0: roll paper end detector disabled 1: roll paper end detector enable 4
	Undefined 5
	Not defined 6
	Not defined 7
	undefined

Note: 1. the item of near-end mounting machines, non-loading machine is invalid.

[Detail] - This command is valid in the parallel / USB.

- simultaneously selecting a plurality of paper without detectors are possible.

· If multiple are effectively selected the detector detects by either detector paper

And it outputs a cut of the signal.

- There may be a delay occurs in the switching of the detector according to the state of the reception buffer. Bit 0 / bits 2 and 3, respectively point to the same detector, if one or the other but 1

It validates the detector. · OFF-LINE switch **OFF** For the signal with or without the detector does not output. - initial value is n = 12 (0Ch).

ESC c 5

Compatible models	BL2-58	SK1-41 / 31/32/21/22/24 SP2-21	/ SP3-21	SD1-31	SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] enable or disable the panel switch Code] <1B>

<63> <35> n [domain] 0 ≦ n ≦ 255

Switches to enable or disable the [Function] panel switch.

n = <xxxxxxx0> B: n = <xxxxxxx1> to enable the panel switch B: To disable the panel switch

It is valid only [Detail] · n least significant bit of.

If you have a panel switch is disabled (except for the POWER button), disable all of the panel switch

To become.

• The initial value, and n = 0.

ESC c 6

Compatible models	SK1-41 / 31/32/21/22/24 SD3-21	/ 22	SK1-21H / 31H SK1-211 / 311

Enable / disable [Name] paper loading [Code]

<1B> <63> <36> n [domain] 0 ≦ n ≦ 255

Switches to enable or disable the paper loading operation in [Function] head down or autoloading.

n = <xxxxxxx0> B: Allow the paper loading (permission of the loading operation) n = <xxxxxxx1> B: prohibition of the paper loading (prohibition of loading operation)

It is valid only [Detail] · n least significant bit of.

• During prohibition to continue the paper error even if the paper is loaded. • The initial value, and n = 0.

[Applicable] · SK1-21 / 31 series: to apply to the later firmware version 1.66.

BELL ESC

RS

Compatible models	SP1-21

[Name] buzzer sounding [Code] <07> or <1B> <1E>

[Function] to 200ms buzzer.

ESC p

Compatible models	SD3-21 / 22

[Name] occurrence of the specified pulse

Code] <1B> <70> m t1 t2 [domain] m =

0,1,48,49

0 ≦ t1 ≦ 255 0 ≦

t2 ≦ 255

Against a connector specified in Function] m, outputs a pulse signal of a specified time t1 and t2

To.

m	Description
0,48	drawer kick-out connector pin 2
1,49	Drawer kick-out connector pin 5

The Detail] on time and t1 × 2 ms, the off time and t2 × 2 ms.

For · t2 <t1, the off-time is t1 × 2 ms.

I-9. Response command

GS a

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] Automatic stearyl - enable or disable the task transmission

Code] <1D> <61> n [domain] 0 ≦ n ≦ 255

Select Function] subject to automatic status transmission status.

BIT	function	number
0	0: Non-select the drawer kick-out connector pin 3 status. 1: Select the drawer kick-out connector pin 3 status.	①
1	0: non-selected the online / offline status 1: Select the online / offline status	②
2	0: Non-select the error status 1: Select the error status	③
3	0: non-selected status sheet detector 1: Select the status of the paper detector	④
Four	Undefined 5	
	Not defined 6	
	Not defined 7	
	undefined	

If the [Detail], any even one status became effective, all of the status at the time of executing this command (4 bytes) sends, after the time the change in the states of the selected status, sends a status.

- Overview, please refer to the "1.5 transmission of the printer". • The initial value, and n = 0.

· SK1-21 / 31 V1.81 or later and, in SK1-24 / 41 V2.26 or later, ESC @ After sending command also set To hold the value.

· BIT0 applies to the drawer kick corresponding model.

Automatic status response, in order to automatically send the error status, drawer kick About the state change of the pin 3 of the connector, not subject to the automatic status.

The first byte (printer information)

BIT	status	number	value
0	unused		0
1	unused		0
2 Note 1	Drawer kick-out connector pin 3 = "L" drawer kick-out connector pin 3 = "H"	①	0 1
3	Online state Offline state	②	0 1
Four	unused		1
Five Note 2	Head Closed / paper cover close head open / paper cover open	②	0 1
6	undefined		-
7	unused		0

Note 1. Apply to the drawer kick corresponding model.

Note 2. Apply to SD3-21 / 22 model.

The second byte (error state)

BIT	status	number	value
0	undefined		-
1	undefined		-
2	undefined		-
3	There is no auto-cutter error occurs auto cutter error	③	0 1
Four	unused		0
Five	Voltage abnormality error without the occurrence voltage abnormality error has occurred	③	0 1
6	Yes enables automatic restoration automatic error recovery possible error occurs without the occurrence	③	0 1
7	unused		0

The third byte (Paper Sensor information)

BIT	status	number	value
There is p	aper in the 0, 1 roll paper end detector No paper on the roll paper end detector	④	0 1
2, 3 There	is paper on the roll paper near-end detector No paper on the roll paper near-end detector	④	0 1
Four	unused		0
Five	undefined		-
6	undefined		-
7	unused		0

4th byte (Paper Sensor information)

BIT	status	number	value
0	If the bezel mode -A: * 0 fixed. In the case of the bezel mode -B * Paper undetected error (paper jam, etc.). * Paper detects an error (paper jam, etc.). If the presenter: * Paper undetected error (paper jam, etc.). * Paper detects an error (paper jam, etc.).	④	0 0 1 0 1
1	If the bezel mode -A: * The sensor does not detect the paper. **Note 2** * Sensor has detected the paper. **Note 2** If the bezel mode -B: * After paper cutting, there is a sheet in the bezel. **Note 1** If the presenter: * After paper cutting, there is a paper presenter detector. **Note 1**	④	0 1 1 1
2, 3, 5 un	defined		0
Four	unused		0
6	0: No paper in the presenter detector **Note 3** 1: paper presenter detector **Note 3**		0 1
7	unused		0

Note: 1. This state is '0' cleared by withdrawing the paper. In addition, until the withdrawal, it discards the print data.

2. a response at the time of command inquiry but, as the automatic status not to respond. Note 3. SK1-31 / 32/21/22 V1.81

or later and, to apply on or after SK1-24 / 41 V2.26.

Sample Code Example]

```
        n = 0;
        buf [n ++] = 0x1d; buf [n
        ++] = 0x61; buf [n ++] =
        0x0F;

// command data transmission
        Ret = WriteFile (hPort, buf, n, & dwSendSize, NULL); // extraction of
the response data (4 bytes)
        Ret = ReadFile (hPort, res, 4, & dwSendSize, NULL);

// response data discrimination Example
        // The first byte if (res [0] &
        0x08) {
                Label1-> Caption = "off-line";}

        else {
                Label1-> Caption = "online";}

        // the second byte if (res [1] &
        0x08) {
                Label2-> Caption = "Auto Cutter";}

        else if (res [1] & 0x20) {
                Label2-> Caption = "voltage error";}

        else if (res [1] & 0x40) {
                Label2-> Caption = "enables automatic restoration error";}

        else {
                Label2-> Caption = "no error";}

        // third byte if (res [2] & 0x03) {

                Label3-> Caption = "piece of paper";}

        else if (res [2] & 0x0C) {
                Label3-> Caption = "near-end";}

        else {
                Label3-> Caption = "No paper error";}

        // fourth byte if (res [3] & 0x40)
        {
                Label4-> Caption = "there is paper in the presenter";}

        else {
                Label4-> Caption = "No paper in the presenter";}
```

GS r

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] stearyl - transmission [Code] <1D>

<72> n [domain] of task n = 1,2,49,50

To send the [Function] specified status.

n = 1,49: sending the status of the paper detector. n = 2,50:

Undefined

Please refer to the [Detail] · Overview "1.5 transmission of the printer".

(N = 1,49)

BIT	Content
0,1	0: paper roll paper near-end detector 1: no paper in the
Note 1 roll	paper near-end detector 2,3
	0: paper roll paper end detector 1: no paper in the
roll	paper end detector 4
	Unused (0) 5
	Not defined 6
	0: No paper in the presenter detector Note 2 1: paper presenter detector Note 2
7	Unused (0)

Note: 1. the item of near-end mounting machines, non-loading machine is set to "0".

Note 2. SK1-31 / 32/21/22 V1.81 or later and, to apply on or after SK1-24 / 41 V2.26.

(N = 2,50)

BIT	Content
0	Unused (0) 1
	Undefined 2
	Undefined 3
	Undefined 4
	Unused (0) 5
	Not defined 6
	Not defined 7
	Unused (0)

Sample Code Example]

```
n = 0;
buf [n ++] = 0x1d; buf [n
++] = 0x72; buf [n ++] =
0x01;

// command data transmission
Ret = WriteFile (hPort, buf, n, & dwSendSize, NULL);

// of the response data retrieval (1 byte)
Ret = ReadFile (hPort, res, 1, & dwSendSize, NULL);

// of response data discrimination
if (res [0] & 0x0C) {
    Label-> Caption = "piece of paper";}

else if (res [0] & 0x03) {
    Label-> Caption = "near-end";} else {


    Label-> Caption = "normal";}
```

GS DLE

Compatible model	SK1-41 / 31/32/21/22/24 SD3-21 / 22	SK1-21H / 31H SK1-211 / 311	

[Name] realtime stearyl - enable or disable the task transmission Code] <1D>
<10> n [domain] n = 0, 1, '0', 1 '

To choose to enable or disable the [Function] real-time status transmission.

n = 0, '0': to invalidate the real-time status transmission. n = 1, '1': an active real-time status transmission.

[Detail] · initial value is n = 0.

• This command is, SK1-31 / 32/21/22 V1.81 or later and, to apply on or after SK1-24 / 41 V2.26.

DLE EOT

Compatible models	SK1-41 / 31/32/21/22/24 SD3-21 / 22		SK1-21H / 31H SK1-211 / 311

[Name] realtime stearyl - transmission [Code] <10> <04> n
[domain] of task 1 ≦ n ≦ 4

Transmitting the [Function] of interest status in real time.

[Detail] · Each status represents the current status. Status of interest is a 1-byte data.

• This command is processed at the time of reception.

· <DLE EOT n> is for processing at the time of reception, into the bit image data <DLE EOT n>
If mixed, processes the <DLE EOT n> as a real-time status command, it should be noted by the user side since it does not process the bit image data. • This command is, SK1-31 / 32/21/22 V1.81 or later and, to apply on or after SK1-24 / 41 V2.26.

(N = 1)

BIT	status	value
0	unused	0
1	unused	0
2 Note 1	Drawer kick-out connector pin 3 = "L" drawer kick-out connector pin 3 = "H"	0 1
3	Online state Offline state	0 1
Four	unused	0
Five	undefined	-
6	undefined	-
7	unused	0

Note 1. Apply to Doroakikku corresponding model.

(N = 2)

BIT	status	value
0	undefined	0
1	undefined	0
2	There generation head open without error head open error	0 1
3	undefined	0
Four	undefined	0
Five	There stop printing by no paper without printing stop by without paper	0 1
6	Yes No error occurred error	0 1
7	undefined	0

(N = 3)

BIT	status	value
0	undefined	0
1	undefined	0
2	Yes occur without the presenter error presenter error	0 1
3	Yes occurrence of no cut error of cut error	0 1
Four	undefined	0
Five	Yes occurrence of no voltage error of voltage error	0 1
6	There is occurrence of no occurrence of the head temperature error head temperature error	0 1
7	undefined	0

(N = 4)

BIT	status	value
0	undefined	0
1	undefined	0
2, 3 There	is paper in the near-end detector No paper in the near-end detector	0 1
Four	undefined	0
Five	No paper in the paper; the paper end detector in the paper end detector	0 1
6	There is paper in the presenter detector no paper in presenter detector	0 1
7	undefined	0

Specifies the error contents of the sample code examples] n = 2

```
n = 0;
// real-time stearate - effectiveness of + Submit
        buf [n ++] = 0x1d; buf [n
        ++] = 0x10; buf [n ++] =
        0x01;

// real-time stearate - + Submit
        buf [n ++] = 0x10; buf [n
        ++] = 0x04; buf [n ++] =
        0x02;

// command data transmission
        Ret = WriteFile (hPort, buf, n, & dwSendSize, NULL);

// of the response data retrieval (1 byte)
        Ret = ReadFile (hPort, res, 1, & dwSendSize, NULL);

// of response data discrimination
        if (res [0] & 0x04) {
                Label-> Caption = "head open";}

        else if (res [0] & 0x60) {
                Label-> Caption = "piece of paper";}

        else {
                Label-> Caption = "normal";}
```

DLE DC4

Compatible models	SD3-21 / 22

[Name] real-time output of the specified pulse Code] <10>
<14> nmt [domain] n = 1

m = 0,1 1 ≦
t ≦ 8

The connector pins specified in [Function] m, and outputs a pulse signal of a specified time t.

m = 0: Doroakikku pin 2 m = 1: Doroakikku
5 pin

[Detail] - This command is valid when the serial interface.

On time as t × 100ms, the off-time and t × 100ms. - already the case in the pulse output, the command is ignored. (**DLE DC4** Or **ESC p** When a running) printer is in the OFFLINE it can not execute the command.

[Note] - in addition to this command also, in the case of receiving the same data columns and command, this command

To the same operation as, it should be noted by the user.

GS E

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

Response of [Name] string

[Code] <1D> <45> n STRING (d1 ... dk) [domain] 1 ≦ n ≦
16

20H ≦ d ≦ 7EH

[Function] the number of digits of STRING information specified in n, DLE STX character string DLE ETX To reply in format.

n: number of digits of STRING
information (operation example) Host

Printer

GS E 04h "ABCD" →→→ (reception)

Reception) ←←← DLE STX "ABCD" DLE ETX

[Detail] · n Definitions range of, Disable command.

• For transmission, "the transmission of 1.5 printer" summary please see.

GS R1

Compatible models	SK1-41 / 31/32/21/22/24 SD3-21 / 22		SK1-21H / 31H SK1-211 / 311

Checking [Name] Printer Status Code] <1D> <52> n

[domain] n = 1 or'1 '

The [Function] printer status, DLE STX status (4 bytes) DLE ETX To reply in format.

4-byte configuration of the printer status is as follows.

(1 byte) 'R': Waiting 'B': in operation 'E': Error

(2 bytes) '0': normal '1': near end '2': a piece of paper

'3': head lever open '4' and later, see the table below

(3 bytes) second byte: '4' failure factors at the time of response. '0': no corresponding '1': voltage abnormality '2': Temperature anomaly '3': paper jam

(4 bytes) '0': Reservations

Response value	Response value Details ('4' and later)
'Four'	Abnormal voltage, temperature abnormality, paper jam error * 1
'Five'	Auto cutter error
'6'	Reservation
'7'	Of paper sampling waiting * 1
'8'	There is paper in the paper detector * 1

* 1. bezel mounting (operation mode B) or the response value of the presenter mounting machine.

Examples [Detail] and operation is as shown below.

host Printer

GS R 01h →→→ (reception

Reception) ←←← DLE STX "E200" <In the case of a piece of paper> DLE ETX

[Application]

In · SK1-21 / 31, the response of failure factors of 3 bytes is applied after V1.98.

V1.98 previous versions is the response value of '0' reservation.

In · SK1-24 / 41, the response of failure factors of 3 bytes is applied after V2.30.

V2.30 previous versions is the response value of '0' reservation.

GS R1

Compatible models	L2-58	SP1-21	SP2-21 / SP3-21 SD1-31

Checking [Name] Printer Status Code] <1D> <52> n

[domain] n = 1 or'1 '

The [Function] printer status, DLE STX status (4 bytes) DLE ETX To reply in format.

4-byte configuration of the printer status is as follows.

(1 byte)	(2 bytes)	(3 bytes)	(4 bytes)
'R': Waiting 'B': in operation 'E': Error	'0': normal '1': the near-end * 1 '2': a piece of paper '3': Paper cover open * 1 '4' and later, see the table below	'0': Reservations	'0': Reservations

Response value	SP1-21	SP2-21 / SP3-21	BL2-58 / SD1-31
'Four'	Voltage and temperature abnormal	Voltage and temperature abnormal	Temperature anomaly
'5' auto-cutter error		Reservation	Reservation
'6'	Reservation	Reservation	Abnormal voltage
'7'	Reservation	Reservation	Reservation
'8'	Reservation	Reservation	Reservation

*** 1. If there is no corresponding sensor does not respond to the error status.**

Examples [Detail] and operation is as shown below.

host Printer

GS R 01h →→→ (reception

Reception) ←←← DLE STX "E200" <In the case of a piece of paper> DLE ETX

GS R2

Compatible models	BL2-58	SD3-21

The [Name] Check the battery power level of the printer [Code] <1D> <52> n [domain] n = 2 or'2 '[Function] The battery, DLE STX status (4 bytes) DLE ETX To respond as.

(1 byte)	(2 bytes)	(3 bytes)	(4 bytes)
'B'	'T'	'0': 75% or more '1': 50% '2': more than 25% '3': 24% or less	'0': Reservations

[Detail] · Zanryouchi is expressed as a measure of the remaining amount value of the battery.

GS R3

Compatible models	SK1-41 / 31/32/21/22/24 SD3-21 / 22	SK1-21H / 31H SK1-211 / 311

[Name] Automatic end response print start and the cutter cutting the

Code] <1D> <52> n [domain] n = 3 or'3 '

The [Function] printer status, DLE STX status (4 bytes) DLE ETX To reply in format.

4-byte configuration of the printer status is as follows.

(1 byte)	(2 bytes)	(3 bytes)	(4 bytes)
' P ': Print start' C ': Disconnection completion' S ': Cutting start' E ':error	- - 'P' and 'C' case of -'0 ': Normal - - 'E' the case of -'2 ': a piece of paper '3': Head Lever Open '4': Voltage and temperature anomaly or bezel / presenter Paper error of '5': Auto cutter error '7': paper withdrawal waiting. * 1	second byte: '4' failure factors at the time of response. '0': no corresponding '1': voltage abnormality '2': Temperature anomaly '3': paper jam	'0': Reservations

* 1. bezel mounting (operation mode B) or the response value of the presenter mounting machine.

After sending the [Detail] · This command GS R3 sends always print data + Full cut command,

Automatically returns a response only once at the timing of the start of printing and cut-off completed. When the migrated-to an error condition, at a timing of processing a print starting and cut command

Automatically error reply, clears the response flag of the print start / cut off complete. • When you use this command, set to "OFFLINE BUSY = OFF".

And operation example is as shown below.

host Printer

GS R 33h →→→ (receiving print data

→→→ (receiving full cut →→→ (reception

Reception) ←←← DLE STX "P000" DLE ETX <print start> reception) ←←← DLE STX "S000" DLE ETX <cutting start> reception) ←←← DLE STX "C000" DLE ETX <cut disconnection completion>

[Application]

In · SK1-21 / 31, to apply to the release version V1.60 or later. In · SK1-24 / 41, to apply to the release version V2.24 or later.

In · SK1-21 / 31, the response of failure factors of 3 bytes is applied after V1.98.

V1.98 previous versions is the response value of '0' reservation.

In · SK1-24 / 41, the response of failure factors of 3 bytes is applied after V2.30.

V2.30 previous versions is the response value of '0' reservation.

GS I

Compatible models	SP1-21	SK1-41 / 31/32/21/22/24 SP2-21	/ SP3-21	SD3-21 / 22 SK	1-21H / 31H
	SK1-211 / 311				

[Name] transmission of the printer ID

[Code] <1D> <49> n

Transmitting the domain] n = 1 ~ 3 or '1' ~ '3' / 65 ~ 67/90 ~ 92,94,99 [Function]

specified printer ID.

Please refer to the [Detail] · Overview "1.5 transmission of the printer".

Send Value

In the case of ① n = 1 ~ 3

N	Type of ID	Transmission value (1 byte, hexadecimal)
1, '1'	Model ID	SP1-21: 31H SK1-24 / 41/31/32/21/22: 32H SP2-21 / SP3-21: 33H SD3-21 / 22: 34H SK1-21H / 31H: 35H SK1-211 / 311: 36H
twenty two'	Type ID	BIT0: 0: 2-byte code without a corresponding 1: 2-byte code corresponding Yes BIT1 ~ 7: 0: Reservations
3, '3'	ROM version ID	By ROM version

In the case of ② n = 65 ~ 67

To send in the header <5F> + string + NULL <00>.

N	Type of ID	Send String
65	ROM version	"V1.00" (In the case of V1.00)
66	Manufacture name	"SANEI"
67	Model name	"SP1-21" (models: the case of SP1-21)

③ In the case of n = 90 ~ 92,94,99 (SD3-21 / 22 series only)

To send in the header <5F> + string + NULL <00>.

N	Type of ID	Submit column
90	BT address	"12345678012"
91	Module version	"1.00" (in the case of V1.00)
92	Device name	"SD321 00000000"
94	Security information	0x00 (1 byte)
99	PIN code	"1234"

ESC s

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24	SD1-31	SD3-21 / 22	SK1-21H / 31H
	SK1-211 / 311					

[Name] transmit printer information Code]

<1B> <73> n [domain] n = 2 ~ 5

[Function] specified by n, the printer information, and transmits along reply format.

Reply Format

Header <FF> + the specified value (n) + reply data N

	Information type	Reply data
2	Model name	Max 32-digit variable length returned by the (terminating NULL) ASC code
3	About 1 fixed length 8 digits	(no termination) Reply by the ASC code
Four	About 2 fixed length 8 digits	(no termination) Reply by the ASC code
Five	Memory switch fixed length of	4 digits (no termination) The binary code, reply until the memory switch (n1 ~ n4).

Please refer to the [Detail] · Overview "1.5 transmission of the printer".

• When n definition outside the scope of, to disable the command.

[Application]

· SK1 Series: Apply to the release version 1.20 or later. · BL2-58:

To apply to the release version 1.40 or later.

· SP1-21: To apply to the release version 1.40 or later.

ESC v

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24	SD1-31	SP3-21	SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311					

[Name] transmission of the current status Code]

<1B> <76>

[Function] to send the current printer status. Please refer to the [Detail] · Overview "1.5 transmission of the printer".

BIT	Status (Active: 1)
0	Detecting the near-end sensor 1
	Detecting the open sensor of the thermal head 2
	Detect the paper sensor 3
	Detecting the temperature abnormality of the thermal head 4
	It detects the abnormal operation of the automatic cutter 5
2, 3	Not installed: 0 fixed. If the bezel mode -A: * 0 fixed. In the case of the bezel mode -B * After the paper cut to, it does not detect the sensor. In the case of the presenter * It has detected the paper off the track error. 6
2, 3	Not installed: 0 fixed. If the bezel mode -A: * It has detected the sensor. If the bezel mode -B: * After the paper cut on, it has detected the sensor. Note 1 In the case of the presenter * After the paper cut on, it has detected the sensor. Note 1
7 Note 3	Entrapment due to GS G mode is activated.

Note: 1. This state is, by withdrawing the paper, will be cleared to '0'. Until the withdrawal, the print data is read and discarded.

Note 2. BIT5,6 of the sensor applies bezel, in Note 3. SK1 series release version V1.40 and later to refer to the reflection sensor that is mounted on the presenter.

[Application] · SK1 Series: Apply to the release version 1.20 or later.

· BL2-58: To apply to the release version 1.40 or later.

· SP1-21: To apply to the release version 1.40 or later.

Sample Code Example]

```
n = 0;
buf [n ++] = 0x1B; buf [n
++] = 0x76;

// command data transmission
    Ret = WriteFile (hPort, buf, n, & dwSendSize, NULL);

// of the response data retrieval (1 byte)
    Ret = ReadFile (hPort, res, 1, & dwSendSize, NULL);

// of response data discrimination
    if (res [0] & 0x10) {
        Label-> Caption = "Cutter";}

    else if (res [0] & 0x08) {
        Label-> Caption = "temperature anomaly";}

    else if (res [0] & 0x02) {
        Label-> Caption = "head open";}

    else if (res [0] & 0x04) {
        Label-> Caption = "piece of paper";}

    else if (res [0] & 0x01) {
        Label-> Caption = "near-end";}

    else {
        Label-> Caption = "normal";}
```

(For model kanji character is equipped with support - you bet.)

FS &

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] Chinese character mode specified

[code] <1C> <26>

[Function] to designate of the Chinese character mode. And [Detail] · JIS

code selection is valid only when.

If the kanji mode is selected, it processed as Kanji codes of all 2 bytes. · In the initial state, the Chinese character mode is canceled. - **FS C** In it is possible to perform the selection of the kanji code system.

FS.

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] Chinese character mode release

[code] <1C> <2E>

[Function] to release the Chinese character mode. And [Detail] · JIS

code selection is valid only when.

• If you release the Chinese character mode, to process all as a single-byte code. · In the initial state, the Chinese character mode is canceled.

FS C

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] selected kanji code system [Code] <1C>

<43> n [domain] 0 ≦ n ≦ 255

Select Function] kanji code system.

n = <xxxxxxx0> B: JIS code n = <xxxxxxx1> B:

Shift-JIS code

[Detail] · initial state, and n = 0.

- the least significant bit of n only be valid.

FS S

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] character spacing set of full-size character [code]

<1C> <53> nl nr [domain] 0 ≦ nl ≦ 127

0 ≦ nr ≦ 127

The setting of the left space amount of [Function] double-byte characters (nl) and the right amount of space (nr). [Detail] · amount of space to be set, is the amount of space for the double-byte characters of standard size.

- according to the character magnification, amount of space is [character magnification × amount of space. · In standard mode / page mode, it is possible to perform an independent set. · If the character magnification, etc. exceeds the maximum amount of space, it replaces the set amount of the maximum value. • This command is only valid for the Kanji character. • The initial value, nl, and nr = 0.

FS!

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

Bulk mode specification [Name] Korean character

[code] <1C> <21> n [domain] 0 ≦ n ≦ 255

[Function] perform a batch specification of double-byte character printing mode of.

BIT	Item Description	function
0	undefined	-
1	undefined	-
2	Double width character	0: cancel 1: specified
3	Double-height character	0: cancel 1: specified
Four	undefined	-
Five	undefined	-
6	undefined	-
7	Underline	0: cancel 1: specified

If you specify both the Detail]-lateral magnification and double height character size is four double angle.

- the number of lines of the underline of the double-byte character is a two-dot pitch. - it is also possible to set other command, to validate the command last processed. • The initial value, and n = 0.

[The specification which depend on the model] SP1-21

· Multiple of the thickness of the underline in the same row, to unify the thickest line.

FS -

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] underline specification and release of the double-byte character

[code] <1C> <2D> n [domain] 0 ≦ n ≦ 255

Setting the [Function] double-byte characters of the underline.

n = <xxxxx000> B: underline 0 dot pitch

|

n = <xxxxx111> B: underline 7 dot pitch

And [Detail] · n the lower 3-bit only valid.

• This command is only valid for double-byte characters. - underline adds character width and with respect to its character space. • Do not additions to the black-and-white inverted character. • The initial value, and n = 0.

[The specification which depend on the model] SP1-21

· Multiple of the thickness of the underline in the same row, to unify the thickest line.

FS W

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] specified and release of the double-byte character size,

quadruple [code] <1C> <57> n [domain] 0 ≦ n ≦ 255

[Function] to designate and release of the double-height, double-width in double-byte characters.

n = <xxxxxxx0> B: n = <xxxxxxx1> to

release B: to specify

And [Detail] · n least significant bit only valid.

• This command is only valid for double-byte characters. • The initial value, and n = 0.

FS 2

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] external character registration

[Code] <1C> <32> c1 c2 d1 ... dk [domain] 0 ≦ d ≦ 255

k = 72

c1, c2 is different by the kanji code system. Kanji code

system	c1	c2
JIS code	c1 = 77H	21H ≦ c2 ≦ 2FH
Shift-JIS code	c1 = ECH	40H ≦ c2 ≦ 4EH

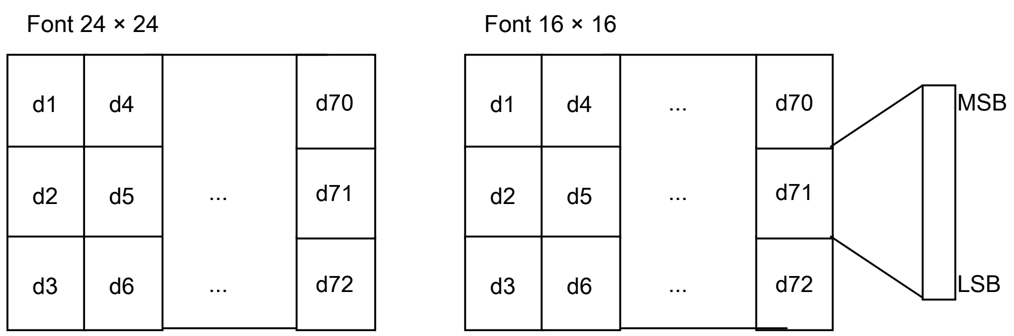
[SK1-211 / 311 series] kanji code

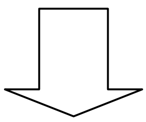
system	c1	c2
JIS code	c1 = 77H	21H ≦ c2 ≦ 7EH
Shift-JIS code	c1 = ECH	40H ≦ c2 ≦ 9EH

The code specified by the [Function] c1, c2, to register as external characters characters. [Detail] · c1 = the first byte, and c2 = the second byte.

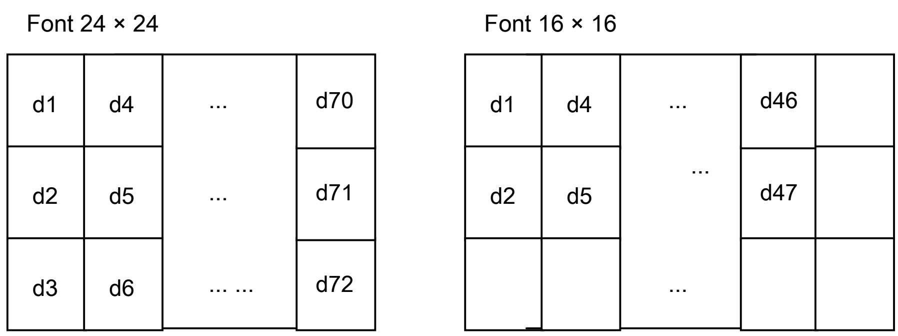
· D is the image data, and a bit to print "1" bits that are not printed "0". · In the character font 16 dots system, and outputs to the lateral width 16 dot vertical width 16 dots.

Registration Image





Character output range



FS Q

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] nonvolatile print image memory, setting the registration mode Code] <1C> <51> n

[Domain] 0 ≦ n ≦ 2 <SK1-41 than>, 0 ≦ n ≦ 4 <SK1-41>

The number specified in [Function] n, to set the registration mode of printing images. [Detail] · after, **FS R** Print contents until the execution is registered in the non-volatile memory.

And registration can be the length of the n = 0,1 is, as a guide to the table below for each model. Data of the print image that exceeds the maximum value that can be registered, truncate. - number of images that can be registered, and two of n = 0,1. If you specify n = 2, the two regions of n = 0, 1

It can be used to register. In SK1-41, you can specify n = 3, 4. . - **ESC J** The paper feed command, such as, not registered as a print image. - during the execution of this command, change of debt / upright printing designation shall not be performed. • This command is, all of the numbers that can be specified by n is, Do not cancel the print image registration mode

And invalid if Kere.

Table (maximum registration line) (Unit: line)

Corresponding model	N = 0,1 maximum	N = 2 maximum	N = 3,4 maximum
BL2-58	1300	2600	- -
SD1-31	1600	3200	- -
SP1 / SP2 / SP3-21	1100	2200	- -
SK1-31 / 32/21/22/24 / 21H / 31H / 211/311	1600	3200	- -
SD3-21 / 22	1800	3200	
SK1-41	1600	3200	2800

And registration of the non-volatile memory, can lead to destruction of the frequently used by you and the non-volatile memory or

Because you, please do not use in such a way that frequently rewritten. - Please do not turn off the power while executing this command absolutely. A danger that the printer failure

Ri you.

FS R

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] nonvolatile print image memory to cancel the registration mode Code] <1C> <52> n

[domain] 0 ≦ n ≦ 2

[Function] print image of the non-volatile memory of the specified number n, to cancel the registration mode. [Detail] · **FS Q** To cancel the execution.

- and later return to the normal operating state.

FS O

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] print image in the nonvolatile memory, sets the print mode Code] <1C> <4F> n

[domain] 0 ≦ n ≦ 2

Setting a print mode for printing images registered in the specified number in the Function] n. [Detail], the non-volatile memory to print images and links that have registered to print.

· N in 倒正 elevational setting of the image that is registered, the same as 倒正 elevational setting of the command execution time

Otherwise, it does not perform a link with the print image. - **ESC J** Paper feed commands, such as does not p erform a link with the print image. - not be done is change of 倒正 standing set this command during execution

.

• This command is, all of the numbers that can be specified by n is, printed image, not to cancel the registration mode

And invalid otherwise.

FS P

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] print image of the non-volatile memory, cancels the print mode Code] <1C> <50> n

[domain] 0 ≦ n ≦ 2

Relative number specified in [Function] n, the print image of the non-volatile memory, to cancel the print mode. [Detail] · **FS O** To release the.

- using this command, it terminates the overprint print image registered in the nonvolatile memory.

FS /

Compatible models	SK1-21 / 22/31/32 SD3-21 / 22	SK1-21H / 31H SK1-211 / 311

[Name] to batch print print image that is registered in the non-volatile memory. [Code] <1C> <2F> n

[domain] 0 ≦ n ≦ 2

For the specified number in the Function] n, the print image registered in the non-volatile memory

To print in bulk.

[Detail] · This command operates after you set the print mode by FS O command.

· SK1 series, is applied to the release version V1.80 or later.

DC3 A

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] Select Border buffer A [Code] <13>

<41>

Select Function] borders buffer A.

[Detail]-ruled line buffer are each independently 2 was (Buffer A, Buffer B) has a built-in,

Selecting a buffer A therein. · Be selected as

the initial value.

DC3 B

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] Select Border buffer B [Code] <13>

<42>

Select Function] borders buffer B.

[Detail]-ruled line buffer are each independently 2 was (Buffer A, Buffer B) has a built-in,

Selecting a buffer B therein. - initial value

selects the buffer A.

DC3 C

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] Clear border buffer [code] <13> <43>

[Function] to clear the contents of the ruled line buffer that has been selected. And

[Detail] clear data are all "0".

DC3 D

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] Write dots designated ruled line buffer [Code] <13> <44> nl nh

[domain] $0 \leqq nl \leqq 255$

$0 \leqq nh \leqq 3$

The dot specified position [Function] Borders buffer "1" writing (black)

Designated position is a [(nh × 256 + nl) × dot pitch.

[Detail] of-ruled line buffer range is set to "0-1023", regardless of the printable area, are selected

To the border buffer "1" write the (black). · Outside the

specified range of data will be ignored.

DC3 L

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21	SD3-21 / 22
	SK1-21H / 31H	SK1-211 / 311				

[Name] Write lines specify borders buffer [Code] <13> <4C> nl nh ml

mh [domain] 0 ≦ nl ≦ 255

0 ≦ nh ≦ 3 0 ≦ ml

≦ 255 0 ≦ mh ≦ 3

In the range of nhnl ~ mhml the [Function] Borders buffer "1" writing (black).

0 ≦ nhnl ≦ mhml ≦ 1023

nhnl = (nh × 256 + nl) × dot pitch. mhml = (mh × 256 +

ml) × dot pitch.

[Detail] of-ruled line buffer range is set to "0-1023", regardless of the printable area, are selected

To the border buffer "1" write the (black). · Outside the

specified range of data will be ignored.

DC3 P

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21	SD3-21 / 22
	SK1-21H / 31H	SK1-211 / 311				

[Name] print execution borders one-dot lines [Code] <13>

<50>

Prints data [Function] print buffer, the 1-dot line in the ruled line buffer selected

For printing.

[Detail] print when there is no data in the buffer, as it is one-dot line printing of the ruled line buffer

It is carried out.

• If the ruled line buffer print mode is prohibited, not the print. In the page mode, it writes the data of the ruled line buffer into the page memory.

DC3 +

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] permission of the borders print mode

[code] <13> <2B>

[Function] to allow the printing mode of the ruled line buffer. [Detail] · permission after, each print command (**CR / LF** Border buffer that is always selected in the etc.)

Adding data for printing. • This command is, **GS L / GS W** Not affected by the printing area commands such as.

Standard mode, the print data next to the printable area of the ruled line buffer, print area

Part out of the is not printed.

Page mode, writes to the area that can be expanded in the page memory, a portion out of the

Not the writing.

- the initial time, the borders print mode is prohibited.

DC3 -

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] prohibition of the borders print mode

[code] <13> <2D>

[Function] to prohibit the printing mode of the ruled line buffer. [Detail] · ban after does not print the data of the ruled line buffer.

I-13. Function or setting command

ESC @

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] Initialization Code] <1B>

<40>

It initializes the [Function] printer.

[Detail] · allocation of user memory is initialized.

And receive buffer is held. Print buffer is cleared. All-various command set is initialized. - to hold the registered data of the non-volatile memory.

DC2 D

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] secure and the opening of the download character registration area

[code] <12> <44> n [domain] 0 ≦ n ≦ 255

[Function] perform a secure and open the download character area.

n = <xxxxxxx0> B: Download character area open n = <xxxxxxx1>

B: Download the character region ensure

When you make the detailed]-open, to plus as user memory free space.

Can not be performed, the open after the download character registration.

• When you make a secured, take out 4560 bytes from the free area of user memory. And securing after is, perform a download character registration. • The initial value, and n = 1.

DC2 G

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] secure and the opening of the registration area of the external character

characters [code] <12> <47> n [domain] 0 ≦ n ≦ 255

[Function] perform a secure and release of external character area.

n = <xxxxxxx0> B: external character area open n =

<xxxxxxx1> B: external character area ensure

When you make the detailed] and opening is plus in the user memory free space.

And opening after the external character can not be performed. Ensure after is, it can be performed external character registration. • When you make a secured, take out 1152 bytes from the free area of user memory. • The initial value, and n = 1.

DC2 ~

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] print density settings [Code]
<12> <7E> n [domain] 50 ≦ n ≦ 160

[Function] sets the print density. [Detail] · n is expressed
as n%.
· When adjusting the print density is doubled by the low thermal paper or the like, and n = 200. • 1 set of each character is because not be done, to validate the last set value. • The initial value, and n = 100. (Model settable in the memory switch obey.) In · SK1-31 / 32, less than release versions V1.33 and 60 ≦ n ≦ 0.99.

DC2!

Compatible models	BL2-58

[Name] Configuration of the double printing
mode Code] <12> <21> n [domain] 0 ≦ n ≦ 255

[Function] for setting and release of the double-printing mode.
n = <xxxxxx00> B: release to n = <xxxxxxx1> B: 2
double-printing and set to n = <xxxxxx1x> B: 2
double-printing • Set
[Detail] · double printing, the printing is line, the mode for printing twice.
· N the lower 2 bits of an effective. • The initial
value, and n = 0.

GS (A

Compatible models	SP1-21	SK1-41 / 31/32/21/22/24 SP2-21	/ SP3-21 SD3-21 / 22	SK1-21H / 31H	
	SK1-211 / 311				

[Name] Run the test print Code] <1D> <28> <41> to
perform the [Function] test print.

To run a test print patterns that are incorporated in the [Detail] printers.

DC2>

Compatible models	SP1-21	SP2-21

[Name] initializes the settings of the print driver
Code] <12> <3E> n [domain] 0 ≦ n ≦ 255

[Function] Sets the dot amount energizes the one pulse. Restore Default
settings [Detail] - print drive.

DC2%

Compatible models	BL2-58	SP1-21	SP2-21

[Name] print drive User Settings [Code] <12>
<25> n [domain] 2 ≦ n ≦ 8

[Function] Sets the dot amount energizes the one pulse. The [Detail] · n × 8 dots, it sets the
maximum applied number of 1 pulse.

· Specified range is set rounding to the minimum value or maximum value. • The initial
value, and n = 8.

[The specification which depend on the model] BL2-58

The initial value of the receipt model, and n = 6.

DC1

Compatible models	SK1-41 / 31/32/21/22/24 SD3-21	/ 22	SK1-21H / 31H SK1	-211 / 311

[Name] software reset [code] <11>

Back to the state it was in when [Function] the power is turned on, to restart.
In [Detail] · USB connection, and about 30 seconds required as the time required for reconnection.

· Received data after the command transmission discards the software reset operation. · SK1 in the series, is applied to
the release version 1.20 or later.

DC2 R

Compatible models	SK1-41 / 31/32/21/22/24 SP2-21 / SP3-21 SD3-21 / 22		SK1-21H / 31H SK1-211 / 311

[Name] Reading Memory Switch [Code] <12> <52>

m [domain] 0 ≦ m ≦ 6

Read the contents of the memory switch specified in the [Function] m, the reply

m = 0: the entire read.

m = 1 ~ 6: read by a number designation of the memory switch.

Read the form, DLE STX of memory switch information (binary) DLE ETX To reply in.

If the [Detail] · one item read only, to specify the m = 1 ... 6. Details of the memory switch

DC2 K checking ...

• For the reply, please refer to the summary "transmission of 1.4 printer". In · SK1-31 / 32, to apply to the release version 1.20 or later.

GS G

Compatible models	SK1-41 / 31/32/21/22/24 SD3-21 / 22		SK1-21H / 31H SK1-211 / 311

[Name] entrapment start and end of the Mode Code] <1D> <47>

n [domain] n = 20h, 21h, 30h, 31h

0 ≦ IDX ≦ 255

For n = 31h, it adds a 4-byte (JOB-ID) in the back. <1D> <47> <31> + ID1 + ID2 + ID3 + ID4

[Function] entrapment specified and the release of the mode.

n = 20h: entrapment release of the mode. n = 21h: entrapment setting of the mode. n = 30h: with JOB-ID, entrapment mode of release n = 31h: with JOB-ID, designation of entrapment mode

The [Detail] · entrapment mode, using the page memory built into the printer, Standard

Print image mode is the save up mode.

· Entrapment When you release the mode, to print all at once the contents of the page memory. - entrapment mode, can not be used in conjunction with page mode.

· 200mm or more write is performed the bulk printing of up to it. However, the mode is not released. · JOB-ID with a function, after you have completed the printing of entrapment, perform the reply of the form.

Reply format: <FF> <13> ID1 ID2 ID3 ID4

• For reply, "the transmission of the 1.5 printer" summary please see. · SK1 in the series, is applied to the release version 1.38 or later.

DC2 u

Compatible models	SD3-21 / 22

[Name] BT device name set in the Code] <12> h <75> h
string NULL [domain] 1 ≦ string ≦ 16

[Function] to register the BT device name of up to a maximum of 16 digits. And
[Detail] · Bluetooth model only valid.
Initial name = string of "XXXXX 000000000" has been registered. (XXXXX is the model name) reflect • This command is effective after power cut. Data of the nonvolatile memory is maintained.

[Note] and registration to the non-volatile memory, can lead to destruction of the frequently used by you and the non-volatile memory
You so, please do not use in any way carry out at any time rewriting. - Please do not turn off the power while executing this command.

DC2 P

Compatible models	SD3-21 / 22

[Name] setting of the PIN code [Code] <12> h
<50> hn string [domain] 1 ≦ n ≦ 16

It has been the number of digits in the string specified by the string = n. (Use string: alphanumeric)

[Function] to register the PIN code of up to a maximum of 16 digits. And
[Detail] · Bluetooth model only valid.
· N represents the number of digits of the string to be registered. · String of initial code = 1234 has been registered. - reflection of the command is valid after power down. Data of the nonvolatile memory is maintained.

[Note] and registration to the non-volatile memory, can lead to destruction of the frequently used by you and the non-volatile memory
You so, please do not use in any way carry out at any time rewriting. - Please do not turn off the power while executing this command.

GS H

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] HRI character printing settings

[Code] <1D> <48> n [domain] 0 ≦ n ≦ 255

[Function] specifies the printing position of the barcode printing when the HRI character.

n = <xxxxxx00> B: HRI not printed characters n = <xxxxxx01> B: printed over the bar code n = <xxxxxx10> B: printed under the bar code n = <xxxxxx11> B: the top and bottom of the bar code print

[Detail] · initial value is n = 0.

· N and the lower two bits only valid.

GS h

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] Set height of the bar code [Code] <1D> <68> n [domain] 1 ≦ n ≦ 255

[Function] perform height setting of the bar code. [Detail] · initial value is n = 162.

GS w

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] setting bar code width Code] <1D> <77> n [domain] 1 ≦ n ≦ 4

[Function] Sets the barcode module width.

n	JAN / UPC of the module width	ITF, CODE39, CODABAR of the module width	
		Naroba	Wide bar
1	2 dot pitch	1 dot pitch	3 dot pitch
2	3 dot pitch	2 dot pitch	5 dot pitch
3	4 dot pitch	3 dot pitch	8 dot pitch
Four	5 dot pitch	4 dot pitch	10 dot pitch

[Detail] · initial value is n = 2.

In the case of · CODE128, the initial value and 2 dot pitch. Setting · CODE128 of the module width is Junsho to JAN / UPC.

GS k

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP2-21 / SP3-21 SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311				

[Name] printing of bar code [code] <1D> <6B> m d1

... dk NUL [domain] 0 ≦ m ≦ 7

d1 ... dk is different with the bar code system.

[Function] Select bar code system, to print the bar code.

m bar	code system 0
	UPC-A 1
	UPC-E 2
	JAN13 3
	JAN8 4
	CODE39 5
	ITF 6
	CODABAR 7
	CODE128 (EAN128)

[Detail] · UPC-A, the data length is 11 bytes, performs internal addition of check digit.

· UPC-E, the data length of 7 bytes, performs internal addition of check digit. · JAN13 the data length is 12 bytes, performs internal addition of check digit. · JAN8 the data length of 7 bytes, performs internal addition of check digit. · CODE39 performs internal addition of a start-stop module. · ITF is the data length and even bytes for internal addition of start / stop module.

· CODE128 is intended to send start module, the bar code data, check di

JIT performs internal addition of the stop module.

However, internal addition of a separator and a check digit for each application identifier by EAN128 is not performed.

For each special character specified by 2 bytes, as follows.

SHIFT -> 7Bh, 53h "{S" CODE A -> 7Bh, 41h "{A"
CIDE B -> 7Bh, 42h "{B" CODE C -> 7Bh, 43h "{C"
FNS 1 -> 7Bh, 31h "{1" FNS 2 -> 7Bh, 32h "{2" FNS
3 -> 7Bh, 33h "{3" FNS 4 -> 7Bh, 34h "{4" '{' -> 7Bh,
7Bh "{{" Start A -> 67h (103) "g" Start B -> 68h (104)
"h" Start C -> 69h (105) "i"

· How to Deploy in the page mode, **ESC L** See.

GS k (GS1 data bar)

Compatible models	SD3-21 / 22 SK1-211 / 311

[Name] GS1 Data Bar printing Code] <1D> <6B> mn d1 ... dk

[domain] 75 ≦ m ≦ 81 (corresponding to GS1 DataBar)

d1 ... dk is different with the bar code system.

[Function] Select bar code system, to print the bar code.

m bar code system
75 GS1 DataBar Omni-directional 76 GS1 DataBar Truncated
77 GS1 DataBar Limited 79 GS1 DataBar Stacked 80 GS1
DataBar Stacked Omni-directional

[Detail] · GS1 DataBar is subject to the following rules.

① n is the number of data, dealing with the n bytes as bar code data from the following data. ② If n is out of definition, to invalidate the command processing.

m	Bar code system	n	d
75	GS1 DataBar Omni-directional	13	48 ≦ d ≦ 57
76	GS1 DataBar Truncated	13	48 ≦ d ≦ 57
77	GS1 DataBar Limited	13	48 ≦ d ≦ 57
79	GS1 DataBar Stacked	13	48 ≦ d ≦ 57
80	GS1 DataBar Stacked Omni-directional	13	48 ≦ d ≦ 57

· GS1 Databar Omni-directional is the data length is 13 bytes, the inside of the check digit

Additional perform.

· GS1 Databar Truncated the data length is 13 bytes, the internal addition of check digit

Carried out.

· GS1 Databar Limited is a data length is 13 bytes, row internal addition of check digit

Nau. 1 byte of data is "0" (48) or, "1" (49) fixed. · GS1 Databar Stacked is the data length is 13 bytes, row internal addition of check digit

Nau.

· GS1 Databar Stacked Omni-directional is the data length is 13 bytes, check de-jitter

Theft of performing the internal added.

(Support in the options corresponding model - you bet.)

GS S

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP3-21	SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311					

[Name] cell change size of the two-dimensional code Code]

<1D> <53> n [domain] 0 ≦ n ≦ 1

[Function] Change the cell size of the two-dimensional code.

n = 0: to the initial value, the cell size of the two-dimensional code. n = 1:

increasing the cell size of the two-dimensional code.

	After the initial value change		
PDF417	2	3	
MicroPDF417	2	3	
DataMatrix	3	Four	
QRCode	3	Four	
MicroQRCode	3	Four	(SK1-21 / 31/41 / 21H / 31H / 211/311 / SD3-21 / 22)

[Detail] · initial value is n = 0.

GS Q

Compatible models	BL2-58	SP1-21	SK1-41 / 31/32/21/22/24 SD1-31		SP3-21	SD3-21 / 22
	SK1-21H / 31H SK1-211 / 311					

[Name] printing of the two-dimensional code Code]

<1D> <51> n Various parameters

Printing the two-dimensional code that is specified in [Function] n.

n = 0: Setting prohibited n =

1: Setting prohibited n = 2:

PDF417 n = 3: MicroPDF417

n = 4: DataMatrix n = 5:

MaxiCode n = 6: QRCode

Apply to SK1-41 / 31/32/21/22 / 21H / 31H / 211/311 / SD3-21 / 22]

n = 7: MicroQRCode (In SK1-31 / 32/21/22 is applied to a release version 1.67 or later.)

※ n subsequent parameters - data, see below.

PDF417

[Code] <1D> <51> 2 Type EncMode ECC_Type ECC_LV Size nl nh Data

Type symbol

0: Sutanda - de 1:

Toranke - Doo

EncMode encode mode

0: automatic optimization diene - de

1: binary diene - de

ECC_LV error correction control level

0 ≦ ECC_LV ≦ 7

Designating one of the combination table Size following column and the number of stages.

Combination table columns and number (X = column, Y = number)

0	X 2: Y 4	6	X 7: Y 15
1	X 2: Y 9	7	X 7: Y 20
2	X 2: Y 15	8	X 12: Y 4
3	X 2: Y 20	9	X 12: Y 9
Four	X 7: Y 4	Ten	X 12: Y 15
Five	X 7: Y 9	11	X 12: Y 20

nl, nh :Data size. (Nl lower byte, nh specify the high-order byte)

1 ≦ nhnl ≦ 448

※ de - maximum value of Tasaizu will vary depending on parameters selected.

Data : De - Takodo.

n = nh * 256 to enter the number of the data specified in + nl.

MicroPDF417

[Code] <1D> <51> 3 Type EncMode Size n Data

Type symbol

0: Sutanda - de

1: Code128 emulation - Tomo - de (no specific industry standard) 2: Code128 emulation - Tomo - de (specific industry standard FNC1 1st) 3: Code128 emulator - Tomodo (specific industry standard FNC1 2nd)

EncMode encode mode - de

0: automatic optimization diene - de

1: binary diene - de

Size

Designating one of the combination table below the columns and the number of stages. Combination table columns and number (X = column, Y = number)

0	X 1: Y 11	8	X 3: Y 26
1	X 1: Y 17	9	X 3: Y 44
2	X 1: Y 28	Ten	X 4: Y 4
3	X 2: Y 8	11	X 4: Y 10
Four	X 2: Y 17	12	X 4: Y 12
Five	X 2: Y 26	13	X 4: Y 26
6	X 3: Y 6	14	X 4: Y 44
7	X 3: Y 12		

n :Data size. (Nl lower byte, nh specify the high-order byte)

1 ≦ nhnl ≦ 150

※ de - maximum value of Tasaizu will vary depending on parameters selected.

Data : De - Takodo.

Enter the number of the data specified by n.

DataMatrix

[Code] <1D> <51> 4 Type Cells / SizeXY nl nh Data

Type symbol

0: square 1:
rectangular

Cells (If the symbol is a square)

10, 18, 22, 26, 32, 40, 48

SizeXY (If the symbol is a rectangle)

0: X = 18, Y = 8 1: X
= 32, Y = 8 2: X = 26,
Y = 12 3: X = 36, Y =
12 4: X = 36, Y = 16
5: X = 48 , Y = 16

nl, nh :Data size. (Nl lower byte, nh specify the high-order byte)

1 ≦ nhnl ≦ 448

※ de - maximum value of Tasaizu will vary depending on parameters selected.

Data : De - Takodo.

n = nh * 256 to enter the number of the data specified in + nl.

MaxiCode

[Code] <1D> <51> 5 Type (OPT SC / CC / PC) n Data

Type symbol

0: Sutanda - de 1: Full ECC

2: array de - data structure

OPT (Type is a case of 2 only)

Be sure to specify one or more ※ BIT0: 1 support
- specify the service class BIT1: 1 country -
specifies the code BIT2: 1 Posutoko - specify the
de

SC (if BIT0 the Type has been specified in the OPT only the case of the two is 1)

Service - up to the specified 3 bytes of the service class the ASCII - numbers. Exit at NULL.

CC (when BIT1 the Type has been specified in the OPT only the case of the two is 1)

Country - co - ascii to a specified 3 bytes of de - numbers. Exit at NULL.

PC (if BIT2 the Type has been specified in the OPT only the case of the two is 1)

Posutoko - ascii to a specified 6 bytes of de - ascii of until the alphanumeric / 9 bytes - end with a number / NULL.

n :Data size. (Nl lower byte, nh specify the high-order byte)

1 ≦ nhnl ≦ 150

※ de - maximum value of Tasaizu will vary depending on parameters selected.

Data: de - Takodo.

Enter the number of the data specified by n.

QRCode

[Code] <1D> <51> 6 Size ECC_LV nl nh Data

Size symbol size

Specifications ① · [BL2-58]

1, 4, 6, 8, 10, 12, 14

Specifications ② · [SP1-21 / SD1-31 / SK1-41 / 31/32/21/22] Note 1

1-14

Specifications ③ · [SD3-21 / 22 / SK1-21H / 31H / SK1-211 / 311]

1-40

ECC_LV ECC (Error Correction Control) level

1: L (7%) 2: M
(15%) 3: Q
(25%) 4: H
(30%)

nl, nh: data size. (Nl lower byte, nh specify the high-order byte)

Specifications ①, ②: 1 ≦ nhnl ≦ 448
specification ③: 1 ≦ nhnl ≦ 7089

※ de - maximum value of Tasaizu will vary depending on parameters selected.

Data: de - Takodo.

n = nh * 256 to enter the number of the data specified in + nl.

Note 1. SP1-21 release version 1.42 or later, is applied to the release version 1.45 or later of SK1-31 / 32/21/22.

MicroQRCode

[Code] <1D> <51> 7 Size ECC_LV n Data

Size symbol size

1, 2, 3, 4

ECC_LV ECC (Error Correction Control) level

1: L (7%) (when Size = 1 is valid only this parameter.) 2: M (15%) 3: Q (25%)

n :Data size.

$1 \leqq n \leqq 35$

※ de - maximum value of Tasaizu will vary depending on parameters selected.

Data: de - Takodo.

Enter the number of the data specified by n.

DC2 L

Compatible models	BL2-58	SK1-41 / 31/32/21/22/24 SD3-22		SK1-21H / 31H SK1-211 / 311

[Name] setting of the label

[Code] <12> <4C> n1 n2 n3 n4 [domain] 1 ≦ n1 ≦

255 Units 1mm] [Unit 2mm (SK1 series, SD3-22 series)]

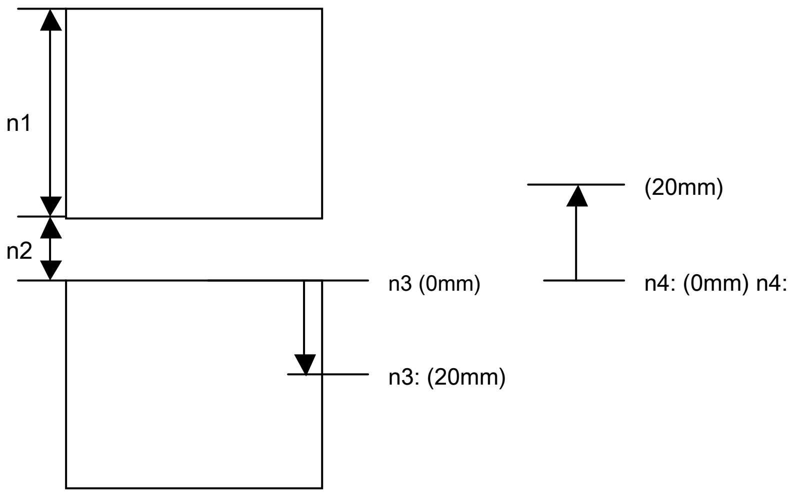
0 ≦ n2 ≦ 20 Units 1mm]

0 ≦ n3 ≦ 20 Units 1mm]

0 ≦ n4 ≦ 20 Units 1mm]

[Function] n1, n2, n3, sets the specified label layout by n4.

n1: to set the paper length. n2: to set the gap length. n3: **DC2 I** After running, it sets the forward paper feed length. n4: Before the start of printing, sets a paper feeding length of the reverse.



[Detail] · The set value is held in the nonvolatile memory.

[The specification which depend on the model] SK1-31 / 32/21/22/41/24/211/311, SD3-22

· N1 parameter is set to a length of n1 × 2 mm.

The maximum length of · n1 parameters will be 350mm. If set to a value greater than the maximum length

The maximum replaced with the length.

- Fear registration to the non-volatile memory, which leads to destruction of the frequently used by you and the non-volatile memory Because there is, please do not use in such a way that frequently rewritten. - Please do not turn off the power while executing this command absolutely. Fear that the printer failure there is.

DC2 I

Compatible models	BL2-58	SK1-41 / 31/32/21/22/24 SD3-22		SK1-21H / 31H SK1-211 / 311

Detection of [Name] marking position [code] <12> <6C>

[Function] perform paper feed until detecting the next marking position.

DC2 B

Compatible models	BL2-58	SK1-41 / 31/32/21/22/24 SD3-22		SK1-21H / 31H SK1-211 / 311

Re-detection of [Name] marking position [code] <12> <42>

[Function] to re-detection of the current marking position. When you run the [Detail] · This command is, please be inserted at the beginning of the printing.

· When executing this command, paper feed in the reverse direction to the detection of the marking position You performed. In this case, the relationship of the marking position detecting sensor and the head holding position, since there is more than the paper (such as mount) is a risk fall out and not in front of the marking position, when using this function, the remainder length please secured.

DC2 mrk

Compatible models	BL2-58	SK1-41 / 31/32/21/22/24 SD3-22		SK1-21H / 31H SK1-211 / 311

[Name] marking threshold settings [Code] <12> <6D> <72> <6B> n
[domain] n = 01h, 05 ~ 30

[Function] performing sensitivity adjustment mark sensor.

n = 1 : Performs twice the paper feeding sheet setting length, optimal for therein thresh To set the level.

n = 05 ~ 30 : Direct, to set the threshold voltage (0.5V ~ 3.0V).

Since the adjustment value by the [Detail] and use the label paper is different, you need to pay attention at the user side.

(SK1 series supports from later V1.40.)

Compatible models	SK1-41 / 31/21/22/24 SK1-21H / 31H	SK1-211 / 311

ESC h

[Name] Selection of presenter Operation Mode [Code] <1B>
<68> n [domain] 0 ≦ n ≦ 7

[Function] to select the mode of operation of the presenter.

n = 0: automatic ejection and recovery (Retraction or Ejection Note 1) Mode n = 1: automatic ejection mode

n = 2: manual release and recovery (Retraction or Ejection Note 1) Mode n = 3: Manual discharge mode n = 4: Forced-discharge operation n = 5: Forced-collection operation n = 6: presenter function OFF n = 7: the presenter function OFF

[Detail] · initial value is n = 0.

- automatic ejection mode, in synchronization with the cutter operation, automatically perform the discharge operation. Automatic recovery mode, in synchronization with the recovery time-out, automatically performing a recovery operation. • Manual discharge mode is, by specifying the user, perform a discharge operation. • Manual recovery mode, by specifying the user, perform the recovery operation. - how to specify the manual mode, **ESC r0** Performed by the instruction. And functions OFF does not perform the operation of the presenter.

- Selection of the recovery operation is performed by a command operation function setting or ESC r3.

ESC r 0

[Name] presenter Manual Operation Code] <1B>
<72> <30> n [domain] 0 ≦ n ≦ 255

[Function] to select the mode of operation of the presenter.

n = <xxxxxxx0> B: Manual and recovery operations (Retraction or Ejection Note 1) N = <xxxxxxx1> B: Manual-discharge operation

[Detail] · **ESC h** At, to function when you have enabled the operating mode.

ESC r 1

[Name] presenter recovery timeout setting Code] <1B> <72> <31>
n [domain] 0 ≦ n ≦ 61

Set [Function] presenter recovery timeout (n × 1 second).
n = 0: to disable the timeout.
[Detail] · initial value is n = 4.

ESC r 3

[Name] Setting presenter Operation Mode [Code] <1B> h
<72> h <33> hn [domain] 0 ≦ n ≦ 3

Select Function] presenter mode of operation.
n = 0: CLAMP / RETRACT n = 1:
CLAMP / EJECT n = 2: CLAMP
ONLY n = 3: CONTINUOUS

[Detail] · initial value is in accordance to the value of the memory switch.
· SK1-31 / 21/22 V1.89 and and apply it to the later SK1-24 / 41 V2.26.

ESC r @

Of [Name] presenter error reset Note 1
[Code] <1B> <72> <40> n [domain] n = 0,
'0'
The state of the [Function] presenter is initialized to clear the error cause.
n = 0, '0': to initialize, to clear the cause of the error.

Note: 1. the recovery operation selection and, ESC r @ co-command, SK1-31 / 21/22 V1.81 or later and applies to SK1-24 / 41 V2.26.

I-18. LED bezel

(SK1-31 / 32/21/22 Series V1.98 or later / SK1-24 series V2.30 supports from later.)

Compatible models	SK1-21 / 22/24/31/32 SK1-211 / 311

GS lnm

[Name] Setting of the emission pattern of the LED with bezel Code]

<1D> <6C> nm [domain] n = 0, 0 ≦ m ≦ 3

[Function] for selecting the operating mode of the light emission pattern of the LED with the bezel.

m = 0: No light emission pattern m =
1: emission pattern 1 m = 2: emission
pattern 2 m = 3: emission pattern 3

[Detail] · initial value, m = 1 To.

[Emission pattern Details of the LED bezel state ○: issuing color lighting of, ●: off, ◎: red lights (0.1sec)

Emission pattern 1 (initial value)

The state of the printer	The light-emitting pattern	Emission color
Off the paper is not in the internal bezel		-
Paper withdrawal waiting due to the mode B	○ ○ ● ●	Green Or Greenery
Paper withdrawal waiting due to the mode A or C	○ ○ ○ ○	Green Or Greenery
Inside feed print or paper sheet exists bezel	○ ○ ○ ○	Green Or Greenery
Printer error	Same emission pattern to a printer error	Red

Light emission pattern 2

The state of the printer	The light-emitting pattern	Emission color
Off the paper is not in the internal bezel		-
Paper in the internal bezel there is ○○	●●	Green Or Greenery
Printer error	Same emission pattern to a printer error	Red

Light emission pattern 3

The state of the printer	Display pattern	Emission color
Bezel inside the paper there is no ○○	●●	Green Or Greenery
Inside feed print or paper sheet exists bezel	○ ○ ◎◎	○ Green Or ◎ Greenery Red
No paper in the internal bezel printer error	Same emission pattern to a printer error	Red

DC2 K (BL2-58)

Compatible models	BL2-58

[Name] Memory Switch Settings [Code] <12>
<4B> m n1 ... n6 [domain] 0 ≦ n1 ... n6 ≦ 255

Depending on the [Function] parameter 'm', and registers of the memory switch.

m = 0: the entire registration (n1 ... n6)

m = 1 ~ 6: to select the number to register. n1 (m = 1)

BIT	Setting items	Settings
0	COMMAND MODE	0: MODE-A 1: MODE-B
1	PAPER FEED	0: OFF 1: ON
2	OFFLINE BUSY	0: ON 1: OFF
3	AUTO POWER OFF	0: OFF 1: ON
4, 5	POWER SWITCH	0: 1sec 1: 2sec 2: 3sec
6	SELECT SENSOR	0: Reflection 1: Transmission
7	MARK DETECTION	0: OFF 1: ON

n2 (m = 2)

BIT	Setting items	Settings
0,, 3	CHARACTER TABLE	0: PC437 1: KATAKANA 2: PC850 3: PC852 4: PC857 5: PC858 6: PC863 7: PC865 8: PC866 9: WPC1252
4,, 6	PRINT DENSITY	0 80% 1: 90% 2: 100% 3: 110% 4: 120% 5: 130% 6: 140% 7: 150%
7	MARK RE-DETECTION	0: OFF 1: ON

n3 (m = 3)

BIT	Setting items	Settings
0,, 2	CHARACTER SET	0: Japan 1: USA 2: Germany 3: England 4: France 5: Spain 6: Italy 7: Sweden
3	PRINT MODE	0: Graphics 1: Character
Four	FONT SIZE	0: 24dot 1: 16dot
Five	UPRIGHT / INVERT	0: Upright 1: Invert
6	PRINT SELECTION	0: Normal Print 1: Double Print
7	NC	undefined

n4 (m = 4)

BIT	Setting items	Settings
0,, 2	BAUD RATE	0: 1200 1: 2400 2: 4800 3: 9600 4: 19200 5: 38400 6: 57600 7: 115200
3	BIT LENGTH	0: 8bit 1: 7bit
4, 5	PARITY	0: Non 1: Non 2: Odd 3: Even
6	BUSY CONTROL	0: RTS / CTS 1: XON / XOFF
7	USB DEVICE CLASS **Note 1**	0: PRINTER 1: SERIAL

Note: 1. to apply to the release version V1.40 or later.

n5 (m = 5)

BIT	Setting items	Settings
0,, 7	Value at Paper Feed	0 ≦ n5 ≦ 255

n6 (m = 6)

BIT	Setting items	Settings
0,, 7	Value at AUTO POWER OFF	1 ≦ n6 ≦ 255

DC2 K (SD1-31)

Compatible models	SD1-31

[Name] Memory Switch Settings [Code] <12>
<4B> m n1 ... n6 [domain] 0 ≦ n1 ... n6 ≦ 255

Depending on the [Function] parameter 'm', and registers of the memory switch.

m = 0: the entire registration (n1 ... n6)

m = 1 ~ 6: to select the number to register.

n1 (m = 1)

BIT	Setting items	Settings
0	COMMAND MODE	0: MODEA 1: MODEB
1	PAPER FEED	0: OFF 1: ON
2	OFFLINE BUSY	0: ON 1: OFF
3-7	NC	undefined

n2 (m = 2)

BIT	Setting items	Settings
0,, 3	CHARACTER TABLE	0: PC437 1: KATAKANA 2: PC850 3: PC852 4: PC857 5: PC858 6: PC863 7: PC865 8: PC866 9: WPC1252
4,, 6	PRINT DENSITY	0 80% 1: 90% 2: 100% 3: 110% 4: 120% 5: 130% 6: 140% 7: 150%
7	NC	undefined

n3 (m = 3)

BIT	Setting items	Settings
0,, 2	CHARACTER SET	0: Japan 1: USA 2: Germany 3: England 4: France 5: Spain 6: Italy 7: Sweden
3	PRINT MODE	0: Graphics 1: Character
Four	FONT SIZE	0: 24dot 1: 16dot
Five	UPRIGHT / INVERT	0: Upright 1: Invert
6	NC	undefined
7	NC	undefined

n4 (m = 4)

BIT	Setting items	Settings
0,, 2	BAUD RATE	0: 1200 1: 2400 2: 4800 3: 9600 4: 19200 5: 38400 6: 57600 7: 115200
3	BIT LENGTH	0: 8bit 1: 7bit
4, 5	PARITY	0: Non 1: Non 2: Odd 3: Even
6	BUSY CONTROL	0: RTS / CTS 1: XON / XOFF
7	NC	

n5 (m = 5)

BIT	Setting items	Settings
0,, 7	Value at Paper Feed	0 ≦ n5 ≦ 255

n6 (m = 6)

BIT	Setting items	Settings
0,, 7	NC	undefined

DC2 K (SP1-21)

Compatible models	SP1-21

[Name] Memory Switch Settings [Code] <12>
<4B> m n1 ... n6 [domain] 0 ≦ n1 ... n6 ≦ 255

Depending on the [Function] parameter 'm', and registers of the memory switch.

m = 0: the entire registration (n1 ... n6)

m = 1 ~ 6: to select the number to register.

n1 (m = 1)

BIT	Setting items	Settings
0	COMMAND MODE	0: MODE-A 1: MODE-B
1	PAPER FEED	0: ON 1: OFF
2	PAGE MODE Note 1	0: ON 1: OFF
3	undefined	- -
Four	undefined	- -
Five	undefined	- -
6	undefined	- -

Note: 1. to apply to the release version V1.40 or later.

n2 (m = 2)

BIT	Setting items	Settings
0,, 3	CHARACTER TABLE Note 1	0: PC437 1: KATAKANA 2: PC850 3: PC852 4: PC857 5: PC858 6: PC863 7: PC865 8: PC866 9: WPC1252 10: PC860
4,, 6	PRINT DENSITY	0 80% 1: 90% 2: 100% 3: 110% 4: 120% 5: 130% 6: 140% 7: 150%
7	undefined	- -

Note: 1. The character set that does not correspond to the disabled.

n3 (m = 3)

BIT	Setting items	Settings
0,, 5 undefined		- -
6	NEAREND SENSOR	0: ON 1: OFF
7	BUZZER	0: ON 1: OFF

n4 (m = 4)

BIT	Setting items	Settings
0,, 7 undefined		- -

n5 (m = 5)

BIT	Setting items	Settings
0,, 7	Value at Paper Feed	0 ≦ n5 ≦ 255

n6 (m = 6)

BIT	Setting items	Settings
0,, 7 undefined		- -

DC2 K (SP2-21)

Compatible models	SP2-21

[Name] Memory Switch Settings [Code] <12>

<4B> m n1 ... n6 [domain] 0 ≦ n1 ... n6 ≦ 255

Depending on the [Function] parameter 'm', and registers of the memory switch.

m = 0: the entire registration (n1 ... n6)

m = 1 ~ 6: to select the number to register.

n1 (m = 1)

BIT	Setting items	Settings
0	undefined	0
1	PAPER FEED	0: OFF 1: ON
2	OFFLINE BUSY	0: ON 1: OFF
3-7 Undefined		0

n2 (m = 2)

BIT	Setting items	Settings
0,, 3	CHARACTER TABLE	0: PC437 1: KATAKANA 2: PC850 3: PC852 4: PC857 5: PC858 6: PC863 7: PC865 8: PC866 9: WPC1252 10: PC860
4,, 6	PRINT DENSITY	0 80% 1: 90% 2: 100% 3: 110% 4: 120% 5: 130% 6: 140% 7: 150%
7	undefined	0

n3 (m = 3)

BIT	Setting items	Settings
0,, 7 undefined		0

n4 (m = 4)

BIT	Setting items	Settings
0,, 2	BAUD RATE	0: 1200 1: 2400 2: 4800 3: 9600 4: 19200 5: 38400 6: 57600 7: 115200
3	BIT LENGTH	0: 8bit 1: 7bit
4, 5	PARITY	0: Non 1: Non 2: Odd 3: Even
6	BUSY CONTROL	0: RTS / CTS 1: XON / XOFF
7	STOP BIT	0: 1-BIT 1: 2-BIT

n5 (m = 5)

BIT	Setting items	Settings
0,, 7	Value at Paper Feed	0 ≦ n5 ≦ 255 [Unit mm]

n6 (m = 6)

BIT	Setting items	Settings
0	USB CLASS	0: Printer Device Class 1: Communication Device Class
1,, 7 undefined		

[Detail] - This command is held even if the power supply is cut off is recorded in the nonvolatile memory.

• If all registration carried out in bulk, and m = 0, followed by the contents of the 6-byte set of n1 ... n6. If you want to set only, one item, it specifies the unique number of m = 1 ... 6, corresponding to that item

There followed. Examples, 12,4B, the m1, n1.

- For details on the setting contents, please refer to the separate "SP1-21 Technical Manual". - written content, it reads when you start the printer.

- Fear registration to the non-volatile memory, which leads to destruction of the frequently used by you and the non-volatile memory

Because there is, please do not use in such a way that frequently rewritten. - Please do not turn off the power while executing this command absolutely. Fear that the printer failure

there is.

DC2 K (SP3-21)

Compatible models	SP3-21

[Name] Memory Switch Settings [Code] <12>

<4B> m n1 ... n6 [domain] 0 ≦ n1 ... n6 ≦ 255

Depending on the [Function] parameter 'm', and registers of the memory switch.

m = 0: the entire registration (n1 ... n6)

m = 1 ~ 6: to select the number to register.

n1 (m = 1)

BIT	Setting items	Settings
0	undefined	0
1	PAPER FEED	0: OFF 1: ON
2	OFFLINE BUSY	0: ON 1: OFF
3-7	Undefined	0

n2 (m = 2)

BIT	Setting items	Settings
0,, 3	CHARACTER TABLE	0: PC437 1: KATAKANA 2: PC850 3: PC852 4: PC857 5: PC858 6: PC863 7: PC865 8: PC866 9: WPC1252 10: PC860
4,, 6	PRINT DENSITY	0 80% 1: 90% 2: 100% 3: 110% 4: 120% 5: 130% 6: 140% 7: 150%
7	undefined	0

n3 (m = 3)

BIT	Setting items	Settings
0,, 7	undefined	0

n4 (m = 4)

BIT	Setting items	Settings
0,, 2	BAUD RATE	0: 1200 1: 2400 2: 4800 3: 9600 4: 19200 5: 38400 6: 57600 7: 115200
3	BIT LENGTH	0: 8bit 1: 7bit
4, 5	PARITY	0: Non 1: Non 2: Odd 3: Even
6	BUSY CONTROL	0: RTS / CTS 1: XON / XOFF
7	STOP BIT	0: 1-BIT 1: 2-BIT

n5 (m = 5)

BIT	Setting items	Settings
0,, 7	Value at Paper Feed	0 ≦ n5 ≦ 255 [Unit mm]

n6 (m = 6)

BIT	Setting items	Settings
0	USB CLASS	0: Printer Device Class 1: Communication Device Class
1,, 7 undefined		

DC2 K (SD3-21 / 22 series)

Compatible models	SD3-21 / 22

[Name] Memory Switch Settings [Code] <12>
<4B> m n1 ... n6 [domain] 0 ≦ n1 ... n6 ≦ 255

Depending on the [Function] parameter 'm', and registers of the memory switch.

m = 0: the entire registration (n1 ... n6)

m = 1 ~ 6: to select the number to register.

n1 (m = 1)

BIT	Setting items	Settings
0	COMMAND MODE	0: MODE-A 1: MODE-B
1	PAPER FEED	0: OFF 1: ON
2	OFFLINE BUSY	0: ON 1: OFF
3-6 Undefined 7		
	MARK DETECTION	0: OFF 1: ON

n2 (m = 2)

BIT	Setting items	Settings
0,, 3	CHARACTER TABLE	0: PC437, 1: KATAKANA 2: PC850, 3: PC852 4: PC857, 5: PC858 6: PC863, 7: PC865 8: PC866, 9: WPC1252 10: PC860 11: WPC1252_2 12: PC864 13: WPC1254 14: WPC1250 15: WPC1251 16: PC864 17: Book 18: PC737
4,, 6	PRINT DENSITY	0 80% 1: 90% 2: 100% 3: 110% 4: 120% 5: 130% 6: 140% 7: 150%
7	undefined	

n3 (m = 3)

BIT	Setting items	Settings
0,, 2	PRINT WIDTH	[SD3-21] [SD3-22] 0: 58/48 0: 58/48 1: 58/54
3,, 5	MECHANISM SPEED	[SD3-21] [SD3-22] 0: 100mm / s 0: 200mm / s 1: 90mm / s 1: 170mm / s 2: 80mm / s 2: 150mm / s 3: 60mm / s 3: 130mm / s 4: 110mm / s
6	SELLECT NEAR-END	0: ON 1: OFF
7	CHARACTER TABLE Note 2	To function as BIT4 of CHARACTER TABLE

n4 (m = 4)

BIT	Setting items	Settings
0,, 2	BAUD RATE	0: 1200 1: 2400 2: 4800 3: 9600 4: 19200 5: 38400 6: 57600 7: 115200
3	BIT LENGTH	0: 8bit 1: 7bit
4, 5	PARITY	0: Non 1: Non 2: Odd 3: Even
6	BUSY CONTROL	0: RTS / CTS 1: XON / XOFF
7	STOP BIT	0: 1-BIT 1: 2-BIT

n5 (m = 5)

BIT	Setting items	Settings
0,, 7	Value at Paper Feed	0 ≦ n5 ≦ 255 [Unit mm]

n6 (m = 6)

BIT	Setting items	Settings
0	USB DEVICE CLASS	0: Printer Device Class 1: Communication Device Class
1-7 Undefined		

DC2 K (SK1-41 / 31/32/21/22/24, SK1-21H / 31H)

Compatible models	SK1-41 / 31/32/21/22/24 SK1-21H / 31H

[Name] Memory Switch Settings [Code] <12>
<4B> m n1 ... n6 [domain] 0 ≦ n1 ... n6 ≦ 255

Depending on the [Function] parameter 'm', and registers of the memory switch.

m = 0: the entire registration (n1 ... n6)

m = 1 ~ 6: to select the number to register.

n1 (m = 1)

BIT	Setting items	Settings
0	COMMAND MODE Note 2	0: MODE-A 1: MODE-B
1	PAPER FEED	0: OFF 1: ON
2	OFFLINE BUSY	0: ON 1: OFF
3, 4	CUT AFTER FEED-SW Note 1	0: NON 1: PARTIAL CUT 2: FULL CUT
Five	Not defined 6	
	SELECT SENSOR	0: Reflection 1: Transmission
7	MARK DETECTION	0: OFF 1: ON

n2 (m = 2)

BIT	Setting items	Settings
0,, 3	CHARACTER TABLE	0: PC437, 1: KATAKANA 2: PC850, 3: PC852 4: PC857, 5: PC858 6: PC863, 7: PC865 8: PC866, 9: WPC1252 10: PC860 11: WPC1252_2 12: PC862 13: WPC1254 Note 4 14: WPC1250 Note 5 15: WPC1251 Note 5 16: PC864 Note 6 17: Reservations Note 6 18: PC737 Note 7
4,, 6	PRINT DENSITY	0 80% 1: 90% 2: 100% 3: 110% 4: 120% 5: 130% 6: 140% 7: 150%
7	MARK RE-DETECTION	0: OFF 1: ON

n3 (m = 3)

BIT	Setting items	Settings
0,, 2	PRINT WIDTH	[SK1-31 / 32 / 31H], [SK1-21 / 22/24 / 21H], [SK1-41] 0: 80/72 undefined 80/72 1: 60/56 60/56 undefined 2: 58/54 58/54 undefined 3: 83/80 undefined 83/80 4: Undefined undefined 112/104
3,, 5	MECHANISM SPEED	[SK1-31 / 32/21/22/24], [SK1-41 / SK1-21 / 31H] 0: 110mm / s 110mm / s 1: 130mm / s 130mm / s 2: 150mm / s 150mm / s 3: 170mm / s undefined Four: 190mm / s undefined Five: 200mm / s undefined
6	SELLECT NEAR-END Note 2	0: ON 1: OFF
7	CHARACTER TABLE Note 6	To function as BIT4 of CHARACTER TABLE

n4 (m = 4)

BIT	Setting items	Settings
0,, 2	BAUD RATE	0: 1200 1: 2400 2: 4800 3: 9600 4: 19200 5: 38400 6: 57600 7: 115200
3	BIT LENGTH	0: 8bit 1: 7bit
4, 5	PARITY	0: Non 1: Non 2: Odd 3: Even
6	BUSY CONTROL	0: RTS / CTS 1: XON / XOFF
7	STOP BIT	0: 1-BIT 1: 2-BIT

n5 (m = 5)

BIT	Setting items	Settings
0,, 7	Value at Paper Feed	0 ≦ n5 ≦ 255 [Unit mm]

n6 (m = 6)

BIT	Setting items	Settings
0	USB DEVICE CLASS Note 2	0: Printer Device Class 1: Communication Device Class
1-3 Undefined		
4, 5	BEZEL MODE	0: NON (NORMAL MODE) 1: BEZEL MODE-A Note 3 2: BEZEL MODE-B Note 3 3: BEZEL MODE-C Note 5
6	CODE128 TYPE	[SK1-31 / 32/21/22/24 / 21H / 31H], undefined [SK1-41] 0: TYPE-A 1: TYPE-B
7	undefined	

[Detail] - This command is held even if the power supply is cut off is recorded in the nonvolatile memory.

• If all registration carried out in bulk, and m = 0, followed by the contents of the 6-byte set of n1 ... n6. Only if you want to set and one of the items, specifies the unique number of m = 1 ... 6, after the item corresponding to it

Continue. Examples, 12,4B, the m1, n1.

- For details on the setting contents, please refer to the separate "Technical Manual". - written content, it reads when you start the printer.

> - Fear registration to the non-volatile memory, which leads to destruction of the frequently used by you and the non-volatile memory
> Because there is, please do not use in such a way that frequently rewritten. - Please do not turn off the power while executing this command absolutely. Fear that the printer failure
> there is.

[Notes]

Note: 1. to apply to the firmware version 1.16 or later. 2. apply to the later firmware version 1.20.

3. to apply to the firmware version 1.30 or later. Note 4. to apply to the later firmware version 1.48. Note 5 to apply to the firmware version 1.70 or later

Applied to Note 6.SK1-21 / 31 firmware version 1.98 or later, setting is as follows in the table.

Applied to Note 7.SK1-21 / 31 firmware version 1.99 or later, setting is as follows in the table.

n3	n2	n2	n2	n2	CHARACTER TABLE
BIT4	BIT3	BIT2	BIT1	BIT0	
0	0	0	0	0	0: PC437
0	0	0	0	1	1: KATAKANA
			...		
1	0	0	1	0	18: PC737

DC2 K (SK1-211 / 311)

Compatible models	SK1-211 / 311

[Name] Memory Switch Settings [Code] <12>
<4B> m n1 ... n6 [domain] 0 ≦ n1 ... n6 ≦ 255

Depending on the [Function] parameter 'm', and registers of the memory switch.

m = 0: the entire registration (n1 ... n6)

m = 1 ~ 6: to select the number to register.

n1 (m = 1)

BIT	Setting items	Settings
0	COMMAND MODE	0: MODE-A 1: MODE-B
1	PAPER FEED	0: OFF 1: ON
2	OFFLINE BUSY	0: ON 1: OFF
3, 4	CUT AFTER FEED-SW	0: NON 1: PARTIAL CUT 2: FULL CUT
Five	Not defined 6	
	SELECT SENSOR	0: Reflection 1: Transmission
7	MARK DETECTION	0: OFF 1: ON

n2 (m = 2)

BIT	Setting items	Settings
0,, 3	CHARACTER TABLE	0: PC437, 1: KATAKANA 2: PC850, 3: PC852 4: PC857, 5: PC858 6: PC863, 7: PC865 8: PC866, 9: WPC1252 10: PC860 11: WPC1252_2 12: PC862 13: WPC1254 14: WPC1250 15: WPC1251 16: PC864 17: Book 18: PC737
4,, 6	PRINT DENSITY	0 80% 1: 90% 2: 100% 3: 110% 4: 120% 5: 130% 6: 140% 7: 150%
7	MARK RE-DETECTION	0: OFF 1: ON

n3 (m = 3)

BIT	Setting items	Settings
0,, 2	PRINT WIDTH	0: 80/72 1: 60/56 2: 58/54 3: 83/80 4: 76/68
3,, 5	MECHANISM SPEED	0: 110mm / s 1: 130mm / s 2: 150mm / s 3: 170mm / s 4: 190mm / s 5: 200mm / s 6: 220mm / s 7: 250mm / s
6	SELLECT NEAR-END	0: ON 1: OFF
7	CHARACTER TABLE	To function as BIT4 of CHARACTER TABLE

n4 (m = 4)

BIT	Setting items	Settings
0,, 2	BAUD RATE	0: 1200 1: 2400 2: 4800 3: 9600 4: 19200 5: 38400 6: 57600 7: 115200
3	BIT LENGTH	0: 8bit 1: 7bit
4, 5	PARITY	0: Non 1: Non 2: Odd 3: Even
6	BUSY CONTROL	0: RTS / CTS 1: XON / XOFF
7	STOP BIT	0: 1-BIT 1: 2-BIT

n5 (m = 5)

BIT	Setting items	Settings
0,, 7	Value at Paper Feed	$0 \leqq n5 \leqq 255$ [Unit mm]

n6 (m = 6)

BIT	Setting items	Settings
0	USB DEVICE CLASS	0: Printer Device Class 1: Communication Device Class
1-3 Undefined		
4, 5	BEZEL MODE	0: NON (NORMAL MODE) 1: BEZEL MODE-A 2: BEZEL MODE-B 3: BEZEL MODE-C
6	Not defined 7	
	undefined	

[Detail] - This command is held even if the power supply is cut off is recorded in the nonvolatile memory.

• If all registration carried out in bulk, and m = 0, followed by the contents of the 6-byte set of n1 ... n6. Only if you want to set and one of the items, specifies the unique number of m = 1 ... 6, after the item corresponding to it

Continue. Examples, 12,4B, the m1, n1.

- For details on the setting contents, please refer to the separate "Technical Manual". - written content, it reads when you start the printer.

- Fear registration to the non-volatile memory, which leads to destruction of the frequently used by you and the non-volatile memory

Because there is, please do not use in such a way that frequently rewritten. - Please do not turn off the power while executing this command absolutely. Fear that the printer failure there is.

· CHARCTER TABLE of how to set far in the following table Ritonaru.

n3	n2	n2	n2	n2	CHARACTER TABLE
BIT4	BIT3	BIT2	BIT1	BIT0	
0	0	0	0	0	0: PC437
0	0	0	0	1	1: KATAKANA
		...			
1	0	0	1		18: PC737

DC2 K 7 (SK1 series in general)

Compatible models	SK1-41 / 31/32/21/22/24 SK1-21H / 31H	SK1-211 / 311

[Name] Memory switch setting of the presenter mode of operation Code] <12> h
<4B> h <07> hn [domain] 0 ≦ n ≦ 3

[Function] to set the presenter mode of operation to the memory switch.

bit	Setting items	Settings
0,1	PRESENTER MODE	0: CLAMP / RETRACT 1: CLAMP / EJECT 2: CLAMP ONLY 3: CONTINUOUS
2-7 Undefined		

[Detail]-setting is retained even if the power supply is cut off is recorded in the nonvolatile memory.
· SK1-31 / 32/21/22 V1.81 and and apply it to the later SK1-24 / 41 V2.26.

DC2 R 7 (SK1 series in general)

Compatible models	SK1-41 / 31/32/21/22/24 SK1-21H / 31H	SK1-211 / 311

[Name] reading of the memory switch of the presenter mode of operation [code]
<12> h <52> h <07> h
[Domain] reads the memory switch of the presenter mode of operation, to reply.
Reply format: DLE STX reply value (1 byte binary) DLE ETX .

For [more information] · reply, "the transmission of the 1.5 printer" summary please see.
· SK1-31 / 32/21/22 V1.81 and and apply it to the later SK1-24 / 41 V2.26.

- Fear registration to the non-volatile memory, which leads to destruction of the frequently used by you and the non-volatile memory
Because there is, please do not use in such a way that frequently rewritten. - Please do not turn off the power while executing this command absolutely. Fear that the printer failure
there is.

DC2 K 8 (SK1 series in general, SD3-21 / 22)

Compatible models	SK1-41 / 31/32/21/22/24 SD3-21 / 22	SK1-21H / 31H SK1-211 / 311

[Name] communication-related setting memory switch settings

[Code] <12> h <4B> h <08> hn [domain] 0 ≦ n ≦ 255

[Function] Setting the communication-related settings in the memory switch.

bit	Setting items	Settings
0,4 undefined		
Five	PRINTING SAVING Note 1	0: Invalid 1: Valid
6	Act. For Driver	0: Invalid 1: Valid
7	CTS AVAILABLE	0: Invalid 1: Valid

[Detail]-setting is retained even if the power supply is cut off is recorded in the nonvolatile memory.

· SK1-31 / 32/21/22 V1.99 and and apply it to the later SK1-24 / 41 V2.31.

[Notes]

Only to apply to the Note 1.SK-21 / 32H.

DC2 R 8 (SK1 series in general, SD3-21 / 22)

Compatible models	SK1-41 / 31/32/21/22/24 SD3-21 / 22	SK1-21H / 31H SK1-211 / 311

[Name] communication-related setting memory switch reading of [code]

<12> h <52> h <08> h

[Domain] communication related settings reads the memory switch, to reply.

Reply format: DLE STX reply value (1 byte binary) DLE ETX .

For [more information] · reply, "the transmission of the 1.5 printer" summary please see.

· SK1-31 / 32/21/22 V1.99 and and apply it to the later SK1-24 / 41 V2.31.

- Fear registration to the non-volatile memory, which leads to destruction of the frequently used by you and the non-volatile memory

Because there is, please do not use in such a way that frequently rewritten. - Please do not turn off the power while executing this command absolutely. Fear that the printer failure there is.

DC2 K 11 (SD3-21 / 22)

Compatible models	SD3-21 / 22

[Name] Memory switch setting of the print mode [Code] <12> h
<4B> h <0B> hn [domain] 0 ≦ n ≦ 255

[Function] to set the printing mode to the memory switch.

bit	Setting items	Settings
0	Default Kanji	0: ON 1: OFF
1	JIS / Shift JIS	0: JIS 1: Shift JIS
2	Default linefeed	0:28 1:32
3	QUALITY MODE ※ 1	0: OFF 1: ON
4,7 undefined		-

※ 1.SD3-22 only the corresponding

[Detail]-setting is retained even if the power supply is cut off is recorded in the nonvolatile memory.

- Fear registration to the non-volatile memory, which leads to destruction of the frequently used by you and the non-volatile memory
Because there is, please do not use in such a way that frequently rewritten. - Please do not turn off the power while executing this command absolutely. Fear that the printer failure
there is.

DC2 R 11 (SD3-21 / 22)

Compatible models	SD3-21 / 22

[Name] communication-related setting memory switch reading of the print mode [Code]
<12> h <52> h <0B> h
[Domain] printing mode reads the memory switch, to reply.

Reply format: DLE STX reply value (1 byte binary) DLE ETX .

For [more information] · reply, "the transmission of the 1.5 printer" summary please see.

DC2 K 12 (SD3-21 / 22)

Compatible models	SD3-21 / 22

[Name] BT mode of the memory switch settings [Code] <12> h
<4B> h <0C> hn [domain] 0 ≦ n ≦ 255

[Function] to set the BT mode in the memory switch.

bit	Setting items	Settings
0,1	BT SECURITH	0: NON 1: SERVICE 2: LINK
2	BT PAIRNG	0: ON 1: OFF
5,7 undefined		-

[Detail]-setting is retained even if the power supply is cut off is recorded in the nonvolatile memory.

> - Fear registration to the non-volatile memory, which leads to destruction of the frequently used by you and the non-volatile memory
> Because there is, please do not use in such a way that frequently rewritten. - Please do not turn off the power while executing this command absolutely. Fear that the printer failure
> there is.

DC2 R 12 (SD3-21 / 22)

Compatible models	SD3-21 / 22

[Name] BT mode memory switch reading of [code] <12> h
<52> h <0C> h
[Domain] read the BT mode memory switch, to reply.

Reply format: DLE STX reply value (1 byte binary) DLE ETX .

For [more information] · reply, "the transmission of the 1.5 printer" summary please see.

DC2 K 11 (SK1-21H / 31H)

Compatible models	SK1-21H / 31H

[Name] Memory switch setting of the print mode [Code] <12> h
<4B> h <0B> hn [domain] 0 ≦ n ≦ 255

[Function] to set the printing mode to the memory switch.

bit	Setting items	Settings
0,2		
3	QUALITY MODE	0: OFF 1: ON
4,7 undefined		-

[Detail]-setting is retained even if the power supply is cut off is recorded in the nonvolatile memory.

- Fear registration to the non-volatile memory, which leads to destruction of the frequently used by you and the non-volatile memory Because there is, please do not use in such a way that frequently rewritten. - Please do not turn off the power while executing this command absolutely. Fear that the printer failure there is.

DC2 R 11 (SK1-21H / 31H)

Compatible models	SK1-21H / 31H

[Name] communication-related setting memory switch reading of the print mode [Code]
<12> h <52> h <0B> h
[Domain] printing mode reads the memory switch, to reply.
Reply format: DLE STX reply value (1 byte binary) DLE ETX .

For [more information] · reply, "the transmission of the 1.5 printer" summary please see.

Compatible models **SP1-21**

ESC R

[Name] selection of international character [code] <1B> <52> n [domain] 0 ≦ n ≦ 7

[Function] below To select the character set of the countries shown in the serial To-option.

n	Country
0	Japan
1	America
2	Germany
3	England
Four	France
Five	Spain
6	Italy
7	Sweden

[Detail] · outside the specified range of data it will be ignored.

• The initial value, and n = 0.

GS a

[Name] Automatic stearyl - Selection of valid or invalid task response / real-time command Code] <1D> <61> n [domain] 0 ≦ n ≦ 3

[Function] to choose to enable or disable the automatic status response.

To choose to enable or disable the real-time command.

n = 0: to disable the automatic status response. n = 1: To enable the automatic status response. n = 2: To disable the real-time command. n = 3: to enable the real-time command.

If the [Detail] · automatic status became effective, at the time of executing this command **GS r (** MODE-B) of the stator

Send vinegar (1 byte), and later will be sent to every time you change the state of the status. • For the reply, please refer to the summary "transmission of 1.4 printer". Printer can not run if there is in the OFF-LINE command. • The initial value, and disable the automatic status / real-time command.

GS r

[Name] Submit Status Code] <1D> <72>
n [domain] 0 ≦ n ≦ 255

[Function] to send the current printer status.
n = <xxxxxxx1> B: to send.
[Detail] · n is valid only the least significant bit.
• For the reply, please refer to the summary "transmission of 1.4 printer". · GS regardless of the presence or absence of a, to send the current status.

DLE EOT 1

[Name] real-time status transmission of [code] <10> <04> n
[domain] n = 1

[Function] to send the current printer status in real time. It is ignored when the [Detail] real-time command is disabled.
When the real-time command is valid, consistent with the image data (DLE ENQ 1) to've
Case, the run to identify this command, it should be noted at the user side. - This command is effective only serial interface.

<u>GS r</u> as well as <u>DLE EOT 1</u> Transmission contents of:

BIT	status	value
0	there is paper on the roll paper end detector	0
	No paper on the roll paper end detector	1
1	Paper cover detector is closed.	0
	Paper cover detector is open.	1
	No second voltage abnormality error	0
	There voltage abnormality error	1
3	No automatic return possible error	0
	There is automatic return possible error	1
4	printers are stationary.	0
	The printer is an operation state.	1
5	fixed value	1
6	fixed value	1
7	fixed value	0

DLE ENQ 1

[Name] real-time status transmission of [code] <10> <05> n

[domain] n = 1

[Function] This command performs discarding without setting.

DC2 p

[Name] Selection of paper out error [Code]

<12> <70> n [domain] 0 ≦ n ≦ 255

[Function] This command performs discarding without setting.

DC2 m

[Name] detection of the mark position [code] <12>

<6D> s nl nh [domain] 0 ≦ s ≦ 255

0 ≦ (n = nh × 256 + nl) ≦ 65535

Follow the instructions in the [Function] s, performs paper feeding of the n × dot pitch] relates mark position.

s = <xxxxxx00> B: to the paper feed to the forward direction exit the marking position. s = <xxxxxx01> B: to the paper feed in the forward direction until the marking position. s = <xxxxxx10> B: Invalid setting s = <xxxxxx11> B: Setting Disabled n: maximum paper feed amount of up to marking detection.

And [Detail] of · s lower two bits only valid.

- This command retrieves the parameters to nh, determines the validity of the settings.

IV. MODE B (BL2-58 / SD1-31) Command Reference

Compatible models	BL2-58	SD1-31

IV-1. Paper feed command

CR

[Name] carriage return [code] <0D>

Prints data [Function] print buffer, the line feed on the basis of the line spacing being set

Carried out.

After [Detail]-execution, the print start position beginning.

- **CR** After **LF** It shall be void. - **LF** After **CR** It

shall be valid.

- line feed amount is the plus the character height and line spacing of at that time.

LF

[Name] line feed [code] <0A>

[Function] function is the same as the CR. However, LF immediately after the CR is ignored.

ESC J

[Name] after printing the forward paper feed

Code] <1B> <4A> n [domain] 0 ≦ n ≦ 255

[Function] forward perform [n × dot pitch] paper feeding.

When there is data in the [Detail] print buffer, a new line feed operation of paper [n × dot pitch]

It is carried out.

When data is not present, it performs a paper feeding [n × dot pitch. • In the page mode, for the specified direction in the page memory for moving forward cursor.

ESC j

[Name] reverse paper feed after printing

[Code] <1B> <6A> n [domain] 0 ≦ n ≦ 255

[Function] backwards perform [n × dot pitch] paper feeding.

When there is data in the [Detail] print buffer, reverse and line feed operation [n × dot pitch]

Perform a paper feed.

When data is not present, it performs the reverse paper feed [n × dot pitch. • In the page mode, for the specified direction in the page memory for moving the opposite direction cursor. - SD1-31 does not support.

Note: 1. After the reverse direction paper feed, please refer to the printing operation after performing always 2mm or more positive direction paper feed.

Note 2: This command is, because it is designed for applications to adjust the head position of the print (margin amount), uses other than its purpose, paper jam

Please note that the cause of.

ESC C

[Name] page length setting Code] <1B>

<43> <00> n [domain] 1 ≦ n ≦ 255

[Function]

To set the number of lines on a page.

[More information], page breaks, **FF** Carried out in.

· Initial value is not set state.

FF

[Name] page break [code]

<0C>

Perform a new page based on the [Function] page length setting.

When there is data in the [Detail] print buffer, it performs the page break operation on a new line operation.

• After this command is executed, the beginning of the line with the next printing start position.

When the page mode, and exit the page mode to print the page memory. • This command is, **ESC C** It is served in until you set the number of lines to ignore.

ESC 2

[Name] line spacing 16 dot specify [Code] <1B> <32>

[Function] the line spacing is set to 16 dots line. [Detail] · initial line space volume follows the setting of the memory switch.

ESC 0

[Name] line spacing 4 dot specified [code] <1B> <30>

[Function] to set the line spacing to 4-dot line. [Detail] · initial line space volume follows the setting of the memory switch.

ESC 3

ESC A

[Name] line spacing settings

[Code] <1B> <33> n or <1B> <41> n [domain] 0 ≦ n ≦ 255

[Function] the line spacing is set to [n × dot line]. [Detail] · initial line space volume follows the setting of the memory switch.

ESC SP

[Name] inter-character space set

[Code] <1B> H + <20> H + n or <27> D + <32> D + n [domain] 0 ≦ n ≦ 255 <0xxxxxxxB>

[Function] the right inter-character space is set to [n × dot line]. And [Detail] · n the lower 7 bits only valid.

• In the case of double-byte characters, to double the right inter-character space that has been set. - maximum value exceeds (127) inter-character space is set rounded to the maximum value. - initial value is n = 0 or 2. According to the setting of the memory switch.

ESC s

[Name] character spacing left and right settings

[Code] <1B> <73> nl nr [domain] 0 ≦ nl ≦ 127

0 ≦ nr ≦ 127

[Function] The space between the left and right character set to [n × dot line. In the case of [Detail] · double-byte characters, to double the inter-character space that has been set.

- character spacing exceeding the maximum value = 127, sets rounded to the maximum value. - initial value is nl = 0, nr = 0 or 2. According to the setting of the memory switch.

ESC U

[Name] for upside-down printing / release

Code] <1B> <55> n [domain]

0 ≦ n ≦ 255

[Function] to designate and release of the upside-down printing.

n = <xxxxxxx0> B: n = <xxxxxxx1> to

release B: to specify

[Detail] · This setting is to designate / cancellation of a row-by-row basis.

· 1 in line, to reflect the last set to the content. - inverted printing, and effective

only to the characters and underline.

- upside-down printing and the character of the paper feed direction in the lower is the mode to be printed in the left-justified. Page mode is performed only set, not reflected in the print content in the page mode. • The initial value is in accordance with the setting of the memory switch.

HT

[Name] horizontal tab movement

[code] <09> H

The [Function] printing position, moves to the next horizontal tab position.

If the Detail] and horizontal tab position is not set, the command is ignored.

Configuring horizontal tab position is carried out by ESC D.

ESC D

[Name] horizontal tab position setting [Code] <1B>

<44> n1 ... nk <00> [domain] 1 ≦ n ≦ 255

0 ≦ k ≦ 32

[Function] Sets the horizontal tab position.

n indicates the number of digits to the set position from the head position of the line. k

represents the number of data to be set.

Horizontal tab positions [Detail]-setting, and [character width × n].

· Character width, character spacing, lateral magnification is also included.

· Programmable tab positions and up to 32. 32 as normal data from the next data when it exceeds
Processing.

· If you set the previous value smaller value during the configuration is recognized as NULL code. - Even if the character width is changed after setting, tab position set does not change. - **ESD D NUL** There when it is entered, clears all tab location. · An initial value, based on the font size selected, to set every 8 characters.

ESC -

[Name] underline Settings [Code] <1B> <2D>

n [domain] 0 ≦ n ≦ 255

[Function] to designate and release of the underline.

n = <xxxxx000> B: underline 0 dot pitch

|

n = <xxxxx111> B: underline 7 dot pitch

And [Detail] · n the lower 3-bit only valid.

- and valid for single-byte / double-byte characters.

- underline adds character width and for the inter-character space.

However, **HT** Not added to the part having been skipped by like.

Under-line, but can be any number of times specified / canceled in a row, the thickness of the maximum value is set to 1 type

select.

• The initial value, and n = 0.

SO

[Name] automatic release with double width expansion specified

[code] <0E>

To specify the [Function] lateral magnification enlarged character in a row. [Detail] · **SO** Input after the, until you enter a new line / release command the single-byte / double-byte characters, the inter-character space

To expand in the transverse direction to 2 times, including a scan. · Released, **DC4, CAN, ESC W0** It is released by the input or line feed operations. - any number of times specified / can be released in a row, it can be mixed with ordinary characters / double width enlarged character.

DC4

[Name] automatic release with double width expansion released

[code] <14>

To cancel the [Function] automatic release with double width expansion specified.

ESC W

[Name] double width larger specify / cancel

Code] <1B> <57> n [domain] 0 ≦ n ≦ 255

To designate and release of [Function] double width enlarged character.

n = <xxxxxxx0> B: n = <xxxxxxx1> to

release B: to specify

And [Detail] · n least significant bits of the only effective

- double width after you specify the enlargement, including the space between characters and the subsequent single-byte / double-byte characters, Yokokata to 2 times

To expand in direction. · Released, **ESC W0** It is

released by.

- any number of times specified / can be released in a row, it can be mixed with ordinary characters / double width enlarged character. •

The double width / double height, and specify both become double-height, double-width characters. • The initial value, and n = 0.

ESC w

[Name] double height specified / release Code]

<1B> <77> n [domain] 0 ≦ n ≦ 255

To designate and release of [Function] double height characters.

n = <xxxxxxx0> B: n = <xxxxxxx1> to

release B: to specify

And [Detail] · n least significant bits of the only effective

· Double-height after you specify the enlargement, including the space between characters and the subsequent single-byte / double-byte characters, Tatekata to 2 times

To expand in direction. · Released, **ESC w0** It is

released by.

- any number of times specified / can be released in a row, it can be mixed with ordinary characters / double height characters. • The

double width / double height, and specify both become double-height, double-width characters. • The initial value, and n = 0.

ESC I

[Name] tone reversal designation / release

Code] <1B> <49> n [domain] 0 ≦ n ≦ 255

[Function] to designate and release of the black-and-white inverted character.

n = <xxxxxxx0> B: n = <xxxxxxx1> to

release B: to specify

And [Detail] · n least significant bits of the only effective

· Specified, on a black background a single-byte / double-byte characters from the white background, to the inverted printing from black on a

white background. - any number of times specified / can be released in one line.

- also including inverted space between characters, but between the lines / under line is not inverted. • If you have a different height of the characters in a row

and black-and-white reversal, line the black-and-white inverted in accordance with the highest character

Nau.

• The initial value, and n = 0.

DC2 Y

[Name] characters Vertical specify / cancel Code]

<1B> <59> n [domain] 0 ≦ n ≦ 2

[Function] to designate and release of character vertical writing.

n = 0: vertical release n = 1: right

vertical writing specify n = 2: left vertical

writing designation

[Detail] and right vertical writing specified, prints by rotating the full-width / half-width characters to the right 90 ℃.

, Left vertical writing specified, prints by rotating the full-width / half-width characters to the left 90 ℃. - the beginning of

the print start position / expansion direction at each designated are the same. • The initial value, and n = 0.

DC2 F

[Name] 16 / 24dot font size selection Code] <12> <46> n

[domain] 0 ≦ n ≦ 255

Make a selection of the [Function] font size.

n = <xxxxxxx0> B: Font B (8 × 16,16 × 16) n = <xxxxxxx1> B:

Font A (12 × 24,24 × 24)

And [Detail] · n least significant bits of the only effective

• After you specify, the subsequent single-byte / double-byte characters, and the selected size. - any number of times can be selected in a row.

The download character as single-byte characters, external character thrive as double-byte characters. • The initial value is in accordance with the setting of the memory switch.

ESC t

[Name] internal character set selection [Code]

<1B> <74> n [domain] 0 ≦ n ≦ 255

[Function] for selecting the internal character set.

n = <xxxxxxx0> B: Ignore n = <xxxxxxx1> B: Ignore n =

<xxxxxx10> B: n = <xxxxxx11> ignore B: katakana

character set

And [Detail] of · n the lower two bits only valid

• The initial value, and n = 3.

ESC R

[Name] International letter designation Code] <1B> <52> n [domain] 0 ≦ n ≦ 7 [Function]

below To select the character set of the countries shown in the serial To-option.

n	Country
0	Japan
1	America
2	Germany
3	England
Four	France
Five	Spain
6	Italy
7	Sweden

[Detail] · outside the specified range of data it will be ignored.

• The initial value is in accordance with the setting of the memory switch.

ESC K FS

&

[Name] Chinese character mode specified

[Code] <1B> <4B> or <1C> <26> [Function] for specifying a Kanji mode. And [Detail] · JIS code selection is valid only when.

• If you are a Chinese character mode is selected, it is treated as a kanji code of all 2 bytes. · In the initial state, the Chinese character mode is canceled. - **FS C** In it is possible to perform the selection of the kanji code system.

ESC H

FS.

[Name] Chinese character mode is canceled

[Code] <1B> <48> or <1C> <2E> [Function] to release the Chinese character mode. And [Detail] · JIS code selection is valid only when.

• If you release the Chinese character mode, to process as all of the single-byte code. · In the initial state, the Chinese character mode is canceled.

FS r

[Name] 1/4 square letter designation

Code] <1C> <72> n [domain] 0 ≦ n ≦ 255

To specify the [Function] 1/4 square character.

n = <xxxxxxx0> B: superscript specified n = <xxxxxxx1> B: character specified subscript

And [Detail] · n least significant bits of the only effective

• 16 dot font size during the selection of, 1/4 square character <2330: '0'> ~ <2339: '9'> only specify.

· 24 dot font size during the selection of, 1/4 square character the following code.

<2321> H ~ <237E> H: alphanumeric <2421> H ~ <247E> H: Hiragana <2521> H ~ <257E> H: an effective only when the katakana · JIS code selection.

FS DC2

[Name] 1/4 square character release

[code] <1C> <12>

To cancel the [Function] 1/4 square character.

Clear the details] · 1/4 square character, thereafter will be printed in double-byte characters.

FS C

[Name] JIS code / Shift JIS code-switching Code] <1C> <43> n [domain] 0 ≦ n ≦ 255

Select Function] kanji code system.

n = <xxxxxxx0> B: JIS code n = <xxxxxxx1> B: Shift-JIS code

And [Detail] · n least significant bit only valid.

• The initial state, and n = 0.

• The JIS code is selected, if you want to enter the single-byte code, enter NULL + <Code>.

(1) download character

The download character is the ability to replace the 1-byte single-byte character code to a character defined by the user. Can define the character size is determined by the font size selection DC2 F.

· 24 If you have selected a dot size is the size of the character area of 24 × 12 dots. • If 16 is selected dot size is the size of the character area of 16 × 8 dots.

In order to print the download characters, download the character selection command **ESC %** Specified, thereafter, please enter the single-byte code that you define.

ESC &

[Name] download character registration

[Code] <1B> <26> c1 c2 [d1 ... dy] 1 ... [d1 ... dy] k [domain] 20H ≦ c1 ≦ c2 ≦ FEH However ≠ 7FH

0 ≦ d ≦ 255

y = 48 (when Font A selection), 16 (at Font B select) k = (c2-c1) +1

(However, if across 7FH is -1.)

To define the download characters in the [Function] specified character code.

c1 = Exit code y = 1 character font data number k =

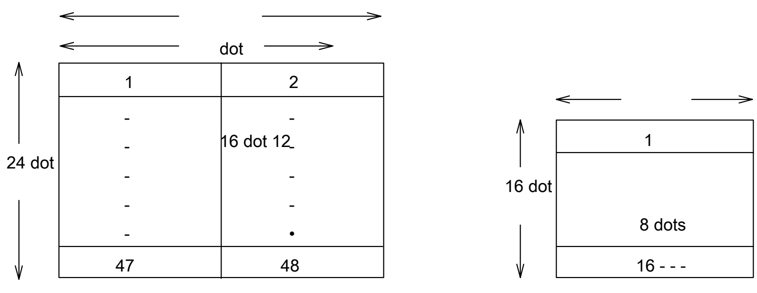
the number of characters that define the character

definition of the start code c2 = character definition

And the case of the [Detail] · one character only of the definition c1 = c2.

· D is the download character of graphic data. • If the last time you specified in the registered code, the process by overwriting. • If you want to validate the registration character fonts, **ESC %** Setting is required. Font data, left → right, is placed horizontally under top →. • 1 byte of font patterns, image LSB / MSB selection **ESC =** In can be changed. • This definition is, but is registered in the user memory, download character area operation **DC2 D** Reserved

There is a need to.



24 dot system

16 dot system

ESC %

[Name] Download Character Selection Code]

<1B> <25> n [domain] 0 ≦ n ≦ 255

[Function] to designate and release the download characters to single-byte character set.

n = <xxxxxxx0> B: n = <xxxxxxx1> to

release B: to specify

And [Detail] · n least significant bit only valid.

• If you cancel the download characters, to specify the internal character set. • If you specify a download character, character set the download characters that are defined

It is designated as.

· Undefined code specifies the internal character set. • The initial value, and n = 0.

DC2 D

[Name] download character area control [Code] <12>

<44> n [domain] 0 ≦ n ≦ 255

Specifying the securing or releasing of the downloaded character region [Function] user memory.

n = <xxxxxxx0> B: n = <xxxxxxx1> to

release B: to ensure

[Detail] · When releasing conducted, the plus as user memory free space.

And release after the download character registration is not carried out.

• When you make a secured, take out 10740 bytes from the free area of user memory. And securing after, the download character registration **ESC &** The perform. - the least significant bit of n only be valid.

- free space in the user memory is the time of less than 10740 bytes, can not be performed is ensured. • The initial value, and n = 1.

(1) external character

External character and is a feature that replaces the 2-byte double-byte character code to a character defined by the user. Can be **defined size of the character, select font size DC2 F It is determined by.**

· 24 If you have selected a dot size is the size of the character area of 24 × 24 dots. • If 16 is selected dot size is the size of the character area of 16 × 16 dots.

In order to print the extended characters, Chinese characters mode FS & Or ESC K Specified, thereafter, please enter a double-byte code that you define.

ESC +

FS 2

[Name] external character definition

[Code] <1B> <2B> c1 c2 [d1 ... dy]

<1C> <32> c1 c2 [d1 ... dy]

[Domain] c1 = 77H

21H ≦ c2 ≦ 7EH 0 ≦

d ≦ 255

y = 72 (when Font A selection), 32 (at Font B select)

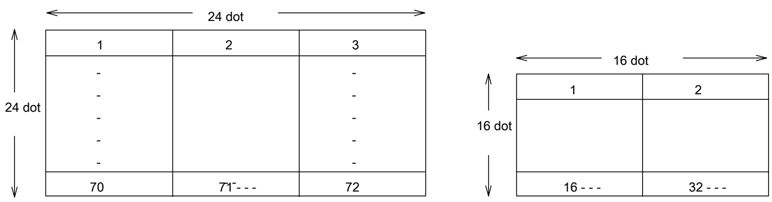
To define the download characters in the [Function] specified character code.

c1 = JIS number of font data of the first byte

c2 = JIS second byte y = 1 character

[Detail] · d is the download character of graphic data.

• If the last time you specified in the registered code, the process by overwriting. • If you want to validate the defined character fonts, **FS &** Or **ESC K** It is necessary to specify the. Font data, left → right, is placed horizontally under top →. • 1 byte of font patterns, image LSB / MSB selection **ESC =** In can be changed. - This definition is registered in the user memory, external character area operation **DC2 G** It is necessary to secure to.



24 dot system

16 dot system

DC2 G

[Name] external character area

control [Code] <12> <47> n [domain]

0 ≦ n ≦ 255

[Function] secure an area external character, or to release.

n = <xxxxxxx0> B: n = <xxxxxxx1> to

release B: to ensure

[Detail] · When releasing conducted, the plus as user memory free space.

- it is not carried out release after the external character definition.

• When you make a secured, take out 6840 bytes from the free area of user memory. - the least significant bit of n only be valid.

- free space in the user memory is when less than 6840 bytes, can not be performed is ensured. • The initial value, and n = 1.

(1) Border

As a ruled line tool as this command emulation, it has a border command. Borders, in order to simplify the creation of such as a table, is the ability to be able to design a horizontal / vertical straight line. Specifically, to save the information about the line in the ruled line buffer, the stored information can be printed by the print execution of the original line feed command or ruled one-dot lines.

As a printing procedure, is as follows mainly.

1. Selection of the border buffer **DC3 A** Or **DC3 B** It is carried out.

2.-border buffer perform a set borders data to the selected border buffer "0" clear (**DC3 C)** To. · Border buffer to the data design (**DC3 V, DC3 D, DC3 L, DC3 F)** To.

3. Allow a border print mode **DC3 +** And, a new line or **DC3 P** To be printed by.

Other, borders can be printed superimposed on the character. However, **ESC J** Or **ESC j** The paper feed command by the paper feed operation except a newline operation is not printed border. Moreover, it not printed as a superposition of borders also Gosutanpu / bit image.

DC3 A

[Name] borders buffer A Select Code <13>
<41>
Select Function] borders buffer A.
[Detail]-ruled line buffer are each independently 2 was (Buffer A, Buffer B) has a built-in,
Selecting a buffer A therein. · Be selected as
the initial value.

DC3 B

[Name] borders buffer B Select Code <13>
<42>
Select Function] borders buffer B.
[Detail]-ruled line buffer are each independently 2 was (Buffer A, Buffer B) has a built-in,
Selecting a buffer B therein. • The initial value has
selected the buffer A.

DC3 V

[Name] borders image write [code] <13> <56> [d1
... dy] [domain] 0 ≦ d ≦ 255

y = 48

[Function] write the image data in the ruled line buffer which has been selected. [Detail] · d is the download character of graphic data.
Image data are arranged horizontally to the left → right. • 1 byte of data patterns, image LSB / MSB selection **ESC =** In can be changed.

DC3 D

[Name] dot set [code] <13> <44> nl nh

[domain] $0 \leqq nl \leqq 255$

$0 \leqq nh \leqq 3$

The [Function] dot specified position of the ruled line buffer which has been selected "1" writing (black)

Designated position is a [(nh × 256 + nl) × dot pitch.

[Detail] of-ruled line buffer range is set to "0-1023" is selected irrespective of the printable area

To the border buffer "1" write the (black). · Outside the

specified range of data will be ignored.

DC3 L

[Name] line set Code] <13> <4C> nl nh ml mh

[domain] $0 \leqq nl \leqq 255$

$0 \leqq nh \leqq 3$ $0 \leqq ml$

$\leqq 255$ $0 \leqq mh \leqq 3$

In the range of nhnl ~ mhml the ruled line buffer [Function] is selected "1" writing (black).

$0 \leqq nhnl \leqq mhml \leqq 1023$

nhnl = (nh × 256 + nl) × dot pitch. mhml = (mh × 256 +

ml) × dot pitch.

[Detail] of-ruled line buffer range is set to "0-1023" is selected irrespective of the printable area

To the border buffer "1" write the (black). · Outside the

specified range of data will be ignored.

DC3 F

[Name] pattern fill Code] <13> <46> n1

n2 [domain] $0 \leqq n1, n2 \leqq 255$

[Function] The writing of one line in the data pattern specified by the selected and are n1 to the ruled line buffer, n2

Carried out.

[Detail] · n1 is 1 byte pattern information, n2 is the second byte pattern information.

Pattern information, as image data, is placed horizontally to the left → right. • 1 byte of data patterns, image

LSB / MSB selection **ESC =** In can be changed.

DC3 +

[Name] borders ON [code] <13>

<2B>

[Function] to allow the printing mode of the ruled line buffer. [Detail] · permission after, each print command (**CR / LF** Data of the ruled line buffer which has been selected in an equal)

Adding to perform printing.

- Border Prints data horizontal printable area of the buffer, outside the area is not printed. It can also affect in-page mode. Initial at the time of, and ruled OFF.

DC3 -

[Name] borders OFF [code] <13>
<2D>
[Function] to prohibit the printing mode of the ruled line buffer. [Detail] · ban after the data of the ruled line buffer is not printed.

DC3 P

[Name row buffer after printing one line printing [Code] <13>
<50>
[Function] performing printing and a line printing of the ruled line buffer the data in the print buffer. If there is no data in the Detail] print buffer, performing one line printing of the ruled line buffer.

• If the ruled line buffer print mode is prohibited, it does not perform printing. • In the page mode, writing of the border buffer in the page memory.

DC3 C

[Name] Clear border buffer [code] <13> <43>

[Function] to clear the contents of the ruled line buffer that has been selected. And
[Detail] clear data are all "0".

DC3 (,)

[Name] borders continuous instruction [code]
<13> <28> ... <29>
[Function] DC3 "(" input after ")" until the input can be an instruction execution borders commands in a row. [Operation] · DC3 other commands are ignored.

- as an input example, DC3 A, DC3 +, when entering continuously DC3 P, continuous instructions as follows

To enter. <13> "(A +
P-)"

ESC V

[Name] bit image specified Code] <1B> <56> nl nh [d1 ...
dk] [domain] 0 ≦ nl ≦ 255

0 ≦ nh ≦ 255 0 ≦
d ≦ 255

[Function] nl, specifies the bit image a specified number of lines in nh.

The number of data of one line	The entire number of data (k)
Refer to the table below.	(Nh × 256 + nl) × 1 number of data lines

Print width	The number of data	Corresponding model
72mm	72	SD1-31
48mm	48	BL2-58

[Detail] · nl, nh indicates the number of vertical lines.

Page mode, to this command and invalid. Specifying the upside-down printing in this command is invalid. Image data, left → right, is placed horizontally under top →. • 1 byte of data patterns, image LSB / MSB selection **ESC =** In can be changed. When a character information printer buffer, overlaid bit image specified character

To be printed. Logo stamp / borders, temporarily stop printing, after printing of bit image, resume the remaining printing. • Example of deployment image, shown in the figure below (print width 48mm).

	D0	D1		D46	D47
1 line 2	D1 * 48 + 0	D1 * 48 + 1	· · · · · · · · · · · · · · · · · ·	D1 * 48 + 46	D1 * 48 + 47
line 3	D2 * 48 + 0	D2 * 48 + 1		D2 * 48 + 46	D2 * 48 + 47
line			- - - - - - -		
nhnl line	Dn * 48 + 0	Dn * 48 + 1	· · · · · ·	Dn * 48 + 46	Dn * 48 + 47

B7 ·········· B0 B0
·········· B7

MSB or LSB LSB or MSB

FS K

[Name] 8 vertical dot bit image specified Code] <1C> <4B>

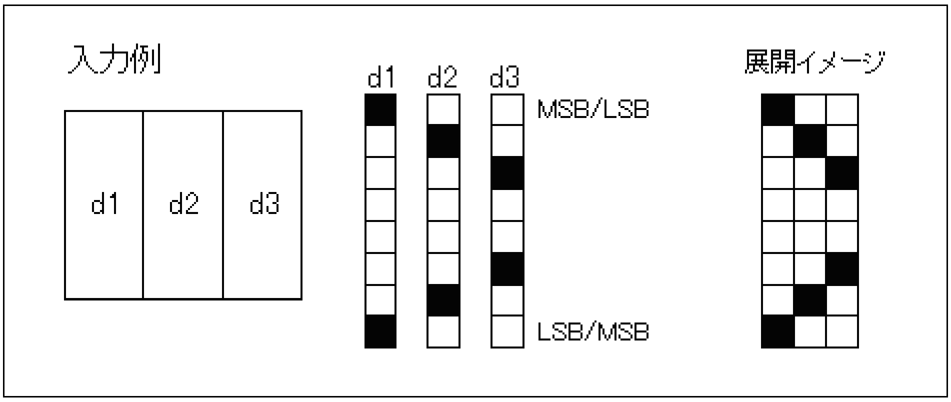
nl nh [d1 ... dk] [domain] 0 ≦ nl ≦ 255

0 ≦ nh ≦ 255 0 ≦

d ≦ 255

[Function] nl, specifies a vertical 8-bit image of the transverse dot number specified in nh. [Detail] · nl, nh indicates the number of lateral dots of the bit image to be printed.

• If you make a dot specified in the printable area outside, it discards the data. The data deployed position, according to the mapping start position of the time. Image data, top → bottom, are arranged perpendicular to the left → right. • 1 byte of data patterns, image LSB / MSB selection **ESC =** In can be changed. And deployment images see the figure below.



(1) logo stamp

The logo stamping, may be registered bit image of rectangle area of any size is a function to print with superimposed with characters. Logo stamp registration **DC2 T** In, please register the bit image of the short-form area. Logo stamp can be registered up to two, logo stamp selection **DC2 S** Logo stamp printed by selecting any of the logo stamp by **DC2 V** Or, it can be printed by the line feed command. It is not possible to print more of the logo stamp at the same time. You can print by superimposing stamp and character.

When you select the logo stamp at the time ruled on, logo stamp to cancel the border printing is printed. When the printing of the logo stamp is finished, the border printing resumes. After you select a logo stamp, command with the securing and releasing of memory areas **DC2 D, DC2 G, DC2)** If you enter, you stop to deselect printing.

DC2 T

[Name] logo stamp registration

[Code] <12> <54> nd yl yh logo stamp data [domain] 0 ≦ n ≦ 255

1 ≦ d ≦ 127

1 ≦ (yh * 256 + yl) ≦ 1023

To the specified number in the Function] n d, yl, registers the logo stamp image specified by yh.

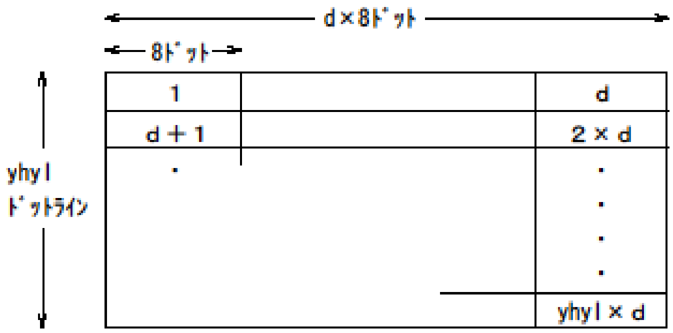
n specifies the logo stamp number. d specifies the number of horizontal direction bytes. yl, yh specifies the number of dots in the vertical direction line.

[More information] logo stamp number is the number of bit image registration is performed.

· Already If you specify a logo stamp number that has been registered, re-erase the previous logo stamp To register.

· D is the lower 7 bits only valid.

· Yh, yl can be specified to 3FFH (1023) dot line. • 1 byte of data patterns, image LSB / MSB selection **ESC =** In can be changed. Logo stamp data registers as shown in the figure below the 8 horizontal dots as 1 byte data.



ロゴスタンプデータ入力順

Data amount of logo stamp data is made to the following formula. Logo stamp the number of registered bytes of = d × (yh × 256 + yl) bytes

• If the number of registered bytes of logo stamp exceeds the remaining memory capacity of user memory, all To ignore the data. Reset, logo stamp will be cleared when the power is turned off.

DC2 S

[Name] logo stamp Selection Code] <12>

<53> nd [domain] 0 ≦ n ≦ 255

0 ≦ d ≦ 255

[Function] Select logo stamp to specify the horizontal print position.

n: logo stamp number d: lateral

printing location

[Detail] · n specifies the logo stamp number registered.

· D specifies the print position in the lateral direction. Specified in the d * 8 dot position. Partial deviated from the printing region by the size of the logo stamp not printed. If the logo stamp specified in n is not registered, an invalid command. - it can not be selected at the same time more than one logo stamp.

• Before printing the logo stamp you choose is completed before, when you enter this command,

To stop the printing of the previous logo stamp, to select the specified parameters.

DC2 V

[Name] logo stamp printing [code] <12>

<56>

To print the logo stamp is [Function] selection to the bottom. If you do not select the

[Advanced] logo stamp and ignored.

DC2 W

[Name] logo stamp kill [code] <12> <57>

[Function] to stop printing the logo stamp is selected. [More information] logo stamp is

ignored if you have not selected.

DC2 U

[Name] logo stamp erasing Code] <12>

<55> n [domain] 0 ≦ n ≦ 255

Clear the logo stamp for the specified number in the Function] n, it releases the memory area used.

n: logo stamp number

[Detail] · When freed, adding the number of bytes that have been released to the user memory.

If you have made a released when the logo stamp has chosen to release to release the selection.

GS k

[Name] print bar codes Code] <1D> <6B> n d1 ... dk
NUL [domain] 2 ≦ n ≦ 7

d1 ... dk is different with the bar code system.

[Function] Select bar code system, to print the bar code.

m	Bar code system
0 1	unused
2	JAN13
3	JAN8
Four	CODE39
Five	ITF
6	CODABAR
7	CODE128 (EAN128)

[Detail] · UPC-A, the data length is 11 bytes, performs internal addition of check digit.
· UPC-E, the data length of 7 bytes, performs internal addition of check digit. · JAN13 the data length is 12 bytes, performs internal addition of check digit. · JAN8 the data length of 7 bytes, performs internal addition of check digit. · CODE39 performs internal addition of a start-stop module. · ITF the data length is an even number of bytes, row internal addition of start / stop module

Cormorant.

· CODE128 is intended to send start module, the bar code data, check
Digit, performs internal addition of the stop module.
However, internal addition of a separator and a check digit for each application identifier by EAN128 is not performed.

For each special character specified by 2 bytes, as follows.

SHIFT -> 7Bh, 53h "{S" CODE A -> 7Bh, 41h "{A"
CIDE B -> 7Bh, 42h "{B" CODE C -> 7Bh, 43h "{C"
FNS 1 -> 7Bh, 31h "{1" FNS 2 -> 7Bh, 32h "{2" FNS
3 -> 7Bh, 33h "{3" FNS 4 -> 7Bh, 34h "{4" '{' -> 7Bh,
7Bh "{{" Start A -> 67h (103) "g" Start B -> 68h (104)
"h" Start C -> 69h (105) "i"

GS w

[Name] setting bar code width Code] <1D>
<77> n1 n2 [domain] 0 ≦ n1 ≦ 2, 0 ≦ n2 ≦ 3

[Function] Sets the barcode module width.

[Unit: dot pitch]

n1	ITF, CODE39, CODABAR of the module width					
	Narrow bar n2	Thick bar				
			0	1	2	3
0	2		Five	6	6	6
1	3		7	8	9	9
2	Four		9	Ten	11	12

[Detail] · initial value is n = 0.
In the case of · CODE128, a two-dot pitch fixed.
- due to the influence of the module width, bar code exceeds the line length is not printed.

GS W

[Name] setting bar code width Code] <1D>
<57> n [domain] 0 ≦ n ≦ 255

[Function] to specify the module width of the UPC / JAN code bar code.
n = <xxxxx001> B: to specify the module width of 1 dot pitch. |

n = <xxxxx111> B: to specify the module width 7 dot pitch.

[Detail] · initial value is n = 3.
In the case of · CODE128, a two-dot pitch fixed.
- due to the influence of the module width, bar code exceeds the line length is not printed. In · BL2-58, to support the release version V1.31 or later. (In BL2-58IR, and later V2.05.)

GS P

[Name] bar code printing position setting [Code]

<1D> <50> n [domain] 0 ≦ n ≦ 2

[Function] sets the printing position of the bar code.

n = 0: print to the left n = 1:

center to print n = 2: printed on

the right

[Detail] · initial value is n = 0.

GS h

[Name] bar code height setting Code] <1D>

<68> n [domain] 1 ≦ n ≦ 255

[Function] perform height setting of the bar code. [Detail] · initial

value is n = 162.

GS H

[Name] bar code data string printed [Code] <1D> <48> n

[domain] 0 ≦ n ≦ 255

[Function] specifies the printing position of the barcode printing when the HRI character.

n = <xxxxxx00> B: Does not print HRI characters. n = <xxxxxx01>

B: printed over the bar code n = <xxxxxx10> B: printed under the

bar code n = <xxxxxx11> B: printing the top and bottom of the bar

code

[Detail] · initial value is n = 0.

· N and the lower two bits only valid.

GS X

[Name] barcode position dot specify [Code] <1D> <58>

nl nh [domain] 0 ≦ nl, nh ≦ 255

[Function] The print start position of the bar code, specified in [(n = nh * 256 + nl) dot pitch. [Detail] · initial value is n = 0.

- due to the influence of the setting, bar code that exceeds the print area is not printed. When · n = 0, **GS P** To reflect the setting of, in ≠ 0 **GS P** Setting is ignored, the position of this command

To reflect the information.

IV- 12 . STX / ETX

(Command to be mounted to the serial interface)

(1) STX / ETX control

STX / ETX control data input by the serial interface, and stored in the input buffer for receiving, whether normally stored data is a control method that can be confirmed by the host. This control can be made in the state in which the input buffer was to "empty".

For communication procedure is as follows.

1. **ENQ** It sends a command to check the status of the printer.
   In particular, to check the state of the input buffer.

2. If the input buffer is "empty" and the normal state, and transmits the STX command.
   STX The command will change the mode to continue to save the data in the input buffer.

3. To send the data. As noted, the maximum size of the input buffer is 8k bytes, the 8k bytes
   Data that exceeds discard read.

Four. The stored data is checked to ensure that it is normal.
   ENQ Send a command, following the status of the printer, 8-bit data of the vertical parity (exclusive <the transmission data, "xor" to calculate>) Check. The initial value of the vertical parity to be responses are determined by a parity bit which is set by the serial interface.

   Odd parity is the initial value = FFH / even parity is an initial value = 0.

Five. **ETX** Send the command, the control is ended.
   The printer is operative to output all the data stored. However, due to no paper during printing, if it becomes OFFLINE temporarily stops until the return to ONLINE.

STX

[Name] the start of the text [code] <02>

[Function] moves to save and go STX / ETX control data into the input buffer for reception. [Detail] · input buffer will be ignored when it is not in a state of "empty".

· Already, when a STX / ETX control does not include this code in the calculation of discarding parity.

ETX

[Name] the end of the text [code] <03>

[Function] Exit STX / ETX control, and outputs the stored data in the input buffer. When not in the [Detail] · STX / ETX control is ignored.

· ETX code is not included in the parity calculation.

ENQ

Response [code] of [Name] printer status <05>

[Function] to 1 byte transmit the printer status. In addition, if it is under the STX / ETX control, perpendicular to the second byte
To send the parity information.

BIT	Function (1: Active state)
0	1: the parity error detection. 1
	1: framing error detection. 2
	1: of mechanical error detection. Detect the hardware abnormality other than the detection-free paper. 3
	1: Detection of paper without error. Four
	0 5
	1: The input buffer is empty. 6
	1: Input buffer is overflowing. 7
	0

When the [Detail] · STX / ETX control to send the vertical parity in the second byte.
· ENQ code is included in the parity calculation.
- This command is for executing at reception buffer development, delayed by the state of the receive buffer
It may occur.
- when responding performs transmission without confirmation of the host state.

DC2>

[Name] fixed division selection Code]

<12> <3E> n [domain] 0 ≦ n ≦ 255

[Function] This command performs discarding without setting.

DC2%

[Name] Number dynamic resolution dot specify

[Code] <12> <25> n [domain] 0 ≦ n ≦ 32

Specifies the number of dots simultaneously energization of [Function] head.

The number of energized dot = become n * 8 dots.

[Detail] · maximum current-carrying number of dots = and 48 dots. If more than 48 dots, replaced by 48.

- current number of dots is related to the current consumption and the printing speed.

- enough to become a large number of dots, the current consumed at one time is large and / printing speed will become faster

You.

· The range of n is set to n = 2 ~ 8, a value exceeding the range on the left-rounding the minimum and maximum values. • The initial value,

and n = 6.

DC2 /

[Name] dynamic resolution temperature, the number of dots

specified Code] <12> <2F> tlh [domain] 0 ≦ t ≦ 255

0 ≦ l ≦ 32 0 ≦

h ≦ 32

Dot Sujo simultaneously performing energization of [Function] head, rows 2 steps specified based on the specified temperature t

Nau.

t: switching temperature (-128 to +127, 2's complement) l: number of energizing dots

below the reference temperature h: becomes equal to the number of energizing dots

below the reference temperature energization number of dots = (l or h) * 8 dots.

[Detail] · maximum current-carrying number of dots, replaced so as not to be greater than the operating number of dots of the recommendation.

- current number of dots is related to the current consumption and the printing speed.

- as it sounds in large numbers, increasing the amount of current to be consumed at one time, printing speed is faster. · DC2% is, l, is set to set the h as the same value, to validate the command set finally. Temperature in order to use the value read by the printer mechanism, the temperature of the external environment

Different from the.

DC2!

[Name] double printing specified / macro print designation

Code] <12> <21> n [domain] 0 ≦ n ≦ 6

[Function] for setting and release of the double-printing mode.

n = 0: 1 double designation of printing n =

1: designation of double print

n = 2: carry out the macro registration / termination of the print data. n = 3: the macro data registered, performing continuous printing twice. n = 4: the macro data registered, performing three consecutive printing. n = 5: the macro data registered, performing four consecutive printing. n = 6: the macro data registered, for 5 consecutive printing.

[Detail] · n = 0, 1 is a mode that specifies printing method.

· Double printing, the printing is line, the mode for printing twice. · N = 2 ~ 6, the macro registration, termination, and mode for execution. And registered the start of the macro, start from always n = 2.

Of-registration end is performed by executing a continuous printing by n = 2 or 3-6. Data size can be registered macro, and up to 2048 bytes. In · SD1-31, it does not support the double-printing mode. • The initial value, and n = 0.

DC2 ~

[Name] print density setting Code]

<12> <7E> n [domain] 0 ≦ n ≦ 255

[Function] sets the print density. [Detail] · n is expressed as n%.

- rated energy that is actually set is set in the range of 50 to 200% set outside this range

It is rounded to within a set.

· When adjusting the print density is doubled by the low thermal paper or the like, and n = 200. • 1 set of each character is because not be done, to validate the last set value. • The initial value is in accordance with the setting of the memory switch.

DC2 r

[Name] stearyl - transmission of Status

Code <12> <72>

[Function] The remaining amount of user memory and sends the ASCII code (hexadecimal six digits). The remaining memory capacity when [Detail] - the power is turned on, 2936 bytes (000B78) is answered.

DC2 e

[Name] Error Status Response Code] <12> <65>

n [domain] 0 ≦ n ≦ 255

[Function] to send the status of the current printer. In addition, automatically varnish during the change of state of the printer

The designation and canceling the transmission of status. n = <xxxxxxx0> B: to cancel the automatic response. n = <xxxxxxx1> B: to specify an automatic response.

BIT	Response (1: Active state)
0	1: of paper without error detection 1
	1: Open cover error detection 2
	1: voltage anomaly detection 3
	1: head temperature anomaly detection 4
	0 5
	1 6
	1 7
	0

When the [Detail] · specified, each time the status changes, sends a status (1 byte).

... initial value and the least significant bit only valid for n, and n = 0.

DC2 q

[Name] execution response request Code]

<12> <71> n [domain] 0 ≦ n ≦ 255

The value set in the [Function] n transmits ORs 50H.

[Detail] · at the end of the print data in addition to the command, if it is confirmed the response value, the printer OFF-LI

Without being NE, which means that it has been to complete the operation successfully. ... initial

value and the lower 4 bits only valid for n, and n = 0.

DC2 v

[Name] voltage response [code]

<12> <76>

The main power supply of the voltage of the [Function] printer, transmitted in ASCII code (decimal, 3 digits).

[Detail]-voltage, if it is 7.6V, responds with a three-digit "7.6".

And voltage remaining capacity of the battery can be confirmed.

DC2 Z

[Name] mode response [code]

<12> <5A>

The Function mode B information dedicated software memory and sends the ASCII code (hexadecimal) 4 bytes. [Detail] · ROMSW1 = 12H, ROMSW2 = time of 34H is responsive and "1234".

DC2 z

[Name] Head Temperature Response

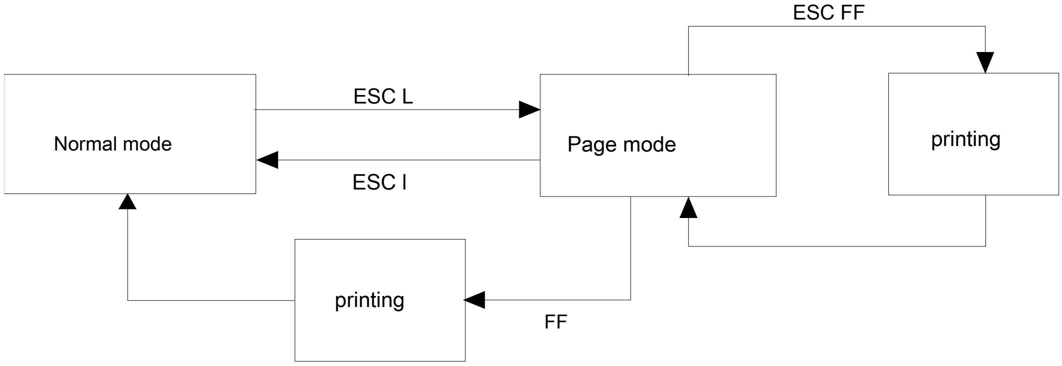
Code] <12> <7A>

The head temperature of the [Function] printer, transmitted in ASCII code (decimal, 3 digits). [Detail]-temperature, if it is -2 ° C., "- 02" and in response, the case of + 25 ° C., responds with "+ 25".

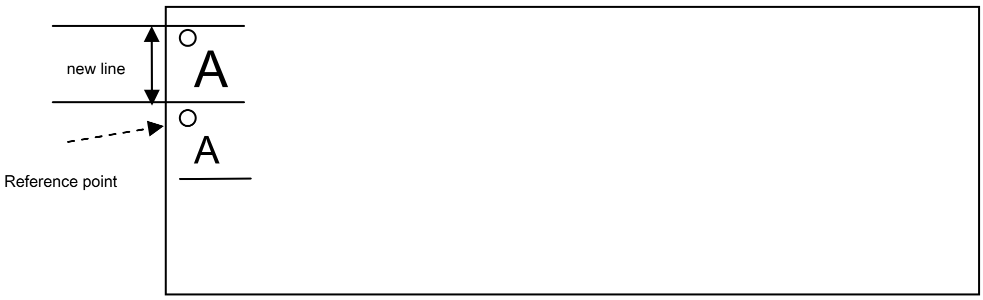
(1) page mode

The page mode, printing instruction (**CR** , **LF** Is also received etc.) does not perform a printing operation, it performs a write to the area of the page memory, **ESC FF** Or **FF** By instruction, it is the operating mode to print all at once the area of the page memory.

Relationship between the page mode and the normal mode is as follows.



Expand the reference point in the page memory is always as a base point to the upper left, deployment in memory operates in the same manner as normal mode. Image is as below.



Data deployed position

ESC L

[Name] page mode selection [code] <1B>

<4C>

[Function] to shift from the normal mode to the page mode. And valid only in the

[Detail] · Normal mode and the beginning of the line.

- **FF** Or **ESC I** By returning to the normal mode. Upside-down printing, is ignored in page mode. - **ESC @** , The process proceeds to the normal mode performs initialization of each mode.

ESC P

[Name] set page size [Code] <1B> <50> nl nh [domain]

8 ≦ (n = nh × 256 + nl) ≦ 1024

[Function] The vertical print area in the page mode is set to [n × dot line. Only effective when processing the [Detail]-page mode.

And horizontal print area is fixed at the maximum printing width. • The first expansion starting position, **ESC T** The upper left relative to the direction specified in. If you specify the-multiple of the page size, to enable the size that is set to last. • The initial value, and n = 1024.

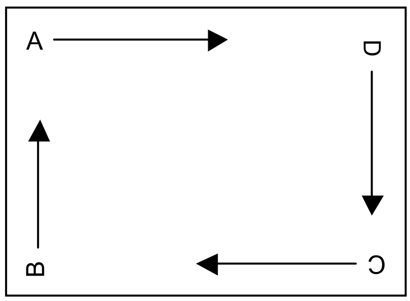
ESC T

[Name] page orientation selected Code]

<1B> <54> n [domain] 0 ≦ n ≦ 3

Selecting a deploy direction and the start of the print data in the [Function page mode.

n	Starting point and developing direction
0	ABCD
1	
2	
3	



Only effective when processing the [Detail]-page mode.

• The initial value, and n = 0.

GS L

[Name] page left and right margin [code]

<1D> <4C> nl nh

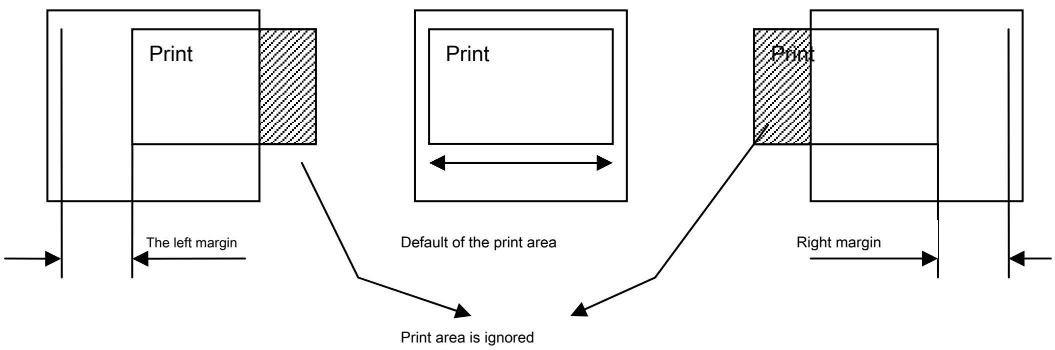
[Domain] -32768 ≦ (n = nh × 256 + nl, 2's complement) ≦ 32767

At the time of printing the execution of the [Function] page mode, to set the left or right margin. Portion exceeding the Detail]-print area is not printed.

· N is a 2's complement negative numbers right margin integer is set as the left margin. • This setting is to be deployed at the time of page printing, if you make more than one set, the last set

To enable the content. • The initial value, and n = 0.

And deployment image is as follows.



ESC FF

[Name] page print [code] <1B>

<0C>

In the [Function] page mode perform batch printing of the printing area. Only effective when processing the [Detail]-page mode.

• After running the contents of the set / printing data of the page mode is retained.

ESC CAN

[Name] page print [code] <1B>

<18>

[Function] page is carried out to clear the memory.

Back to the initial position of the expansion starting position of the [Detail]-page mode, to erase the page memory.

ESC l

[Name] page mode end [code] <1B> <6C>

To exit the Function] page menu mode.

Print data and [Detail] and content that has also been set before the page mode to be completed to hold.

ESC #

[Name] overlay mode selection Code] <1B>

<23> n [domain] 0 ≦ n ≦ 255

Logic processing OR or XOR to print dots [Function] printing position such as characters and graphics overlap

specify.

n = <xxxxxxx0> B: the OR process. n =

<xxxxxxx1> B: the XOR processing.

And [Detail] · n least significant bits of the only effective

• The initial value, and n = 0.

ESC =

[Name] Image LSB / MSB selection Code] <1B>

<30> n [domain] 0 ≦ n ≦ 255

Specifying the image data LSB / MSB of 1 byte in the [Function graph Fick command.

n = <xxxxxxx0> B: the LSB. n = <xxxxxxx1>

B: the MSB.

And [Detail] · n least significant bits of the only effective

· LSB is specified, place the horizontal direction BIT0,1, to ... 7 order of, from the top of the vertical direction B0,1, ... 7 and

Deploy.

· MSB is specified, place the horizontal direction BIT7,6, ... in the order of 0, from the top of the vertical direction B7,6, ... 0 and

Deploy.

• The initial value, and n = 0.

DC2 p

[Name] Selection of paper out error [Code]

<12> <70> n [domain] 0 ≦ n ≦ 255

[Function] This command performs discarding without setting.

DC2 m

[Name] mark position detection [Code] <12>

<6D> s nl nh [domain] $0 \leqq s \leqq 255$

$0 \leqq (n = nh \times 256 + nl) \leqq 65535$

Follow the instructions in the [Function] s, performs paper feeding of the n × dot pitch] relates mark position.

s = <xxxxxx00> B: to the paper feed to the forward direction exit the marking position. s = <xxxxxx01> B: to the paper feed in the forward direction until the marking position. s = <xxxxxx10> B: to the paper feed until in the opposite direction exit the marking position. s = <xxxxxx11> B: to the paper feed in the opposite direction to the marking position.

n: The maximum paper feed amount of up to marking detection.

And [Detail] of · s lower two bits only valid.

- In SD1-31, it does not support this command.

CAN

[Name] Cancel [code] <18>

[Function] carry out the clearing of the print buffer. After the [Detail] and execution, and print starting position the beginning of the line.

ESC @

[Name] Reset [code] <1B> <40>

It initializes the [Function] printer.

[Detail] · allocation of user memory is initialized.

And receive buffer is held. Print buffer is cleared.

- borders except buffer, various command settings are all initialized. Data of the non-volatile memory holds.

ESC S

[Name] set the operating mode Code] <1B> <53>

romsw1 romsw2 [domain] 0 ≦ romsw1,2 ≦ 255

Setting the soft memory of [Function] MODE-B.

ROMSW1

BIT	Setting items	Settings
0,, 3	CHARACTER SET	0: Japan 1: USA 2: Germany 3: England 4: France 5: Spain 6: Italy 7: Sweden
Four	PRINT MODE	0: Graphics 1: Character
Five	FONT SIZE	0: 24dot 1: 16dot
6	NC	
7	PRINT SELECTION **Note 1**	0: Normal Print 1: Double Print

Note In 1.SD1-31, it does not support.

romsw2

BIT	Setting items	Settings
0,, 1	BAUD RATE	0: 9600 1: 19200 2: 4800 3: 2400
2	BIT LENGTH	0: 8bit 1: 7bit
3, 4	PARITY	0: Non 1: Non 2: Odd 3: Even
Five	BUSY CONTROL	0: RTS / CTS 1: XON / XOFF
6	UPRIGHT / INVERT	0: UPRIGHT 1: INVERT
7	NC	undefined

[Detail] - This command is held even if the power supply is cut off is recorded in the nonvolatile memory.

• The initial value is in accordance with the contents of the memory switch.

- For details on the setting contents, please refer to the separate "Technical Manual".

- Fear registration to the non-volatile memory, which leads to destruction of the frequently used by you and the non-volatile memory

Because there is, please do not use in such a way that frequently rewritten. - Please do not turn off the power while executing this command absolutely. Fear that the printer failure

there is.

DC2 y

[Name] set the auto power-off timer [code] <12> <79> n [domain] 0 ≦ n ≦ 255

Set the ON / OFF of the [Function] auto power off, when enabled to set the detection time (min).

n = 0: to disable the auto power-off (OFF).

n = 1 ~ 255: to enable the auto power off (ON), it is set to n minutes.

[Detail] - This command is held even if the power supply is cut off is recorded in the nonvolatile memory.

• The initial value is in accordance with the contents of the memory switch. - SD1-31 does not support this command.

> - Fear registration to the non-volatile memory, which leads to destruction of the frequently used by you and the non-volatile memory Because there is, please do not use in such a way that frequently rewritten. - Please do not turn off the power while executing this command absolutely. Fear that the printer failure there is.

DC2 x

[Name] power off [code] <12> <78>

n

To power down the [Function] printer. [Detail] · SD1-31 does not support this command.

IV-17. label

(In the label model, support you. For more information on command, please refer to the MODE-A.)

DC2 L DC2

I DC2 B

DC2 mrk

MEMO

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